

The Scallop Theorem

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Abstract

In the fluid dynamics of motility, the scallop theorem emerges as the fundamental constraint that arises when generating net forces and flows using a time-periodic deformation at low Reynolds numbers. In this essay, we provide a detailed mathematical exposition of the scallop theorem and its profound consequences. After a summary of the necessary fluid dynamics background, we discuss a rigorous formulation and proof of the theorem. We then review three strategies that are often employed to evade the constraints of the theorem. For each of the strategies, we consider a simple model consisting of a dumbbell swimmer and solve for its swimming speed asymptotically, in the limit of small amplitude deformations. Where possible, we compare the theoretical predictions of classical models with the results of modern experiments.

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1 Introduction

Often attributed to Edward Purcell [1], the scallop theorem was first understood in the context of the locomotion of microorganisms. Despite its significant consequences, the theorem admits a simple formulation. It states that in the absence of inertia, a periodic deformation that looks the same when played in reverse, a so called reciprocal deformation, cannot lead to a net force or flow. It was well-known in the 1950s that the strategies employed by macroscopic organisms for locomotion, such as the flapping of wings, relied on inertia and would be futile in the low-Reynolds number world, inhabited by a plethora of microorganisms. It was G. I. Taylor who first asked if it would be possible for a body to propel itself in the absence of inertia and his subsequent seminal paper [2] is often marked as the genesis of research in the fluid mechanics of locomotion. Indeed, Taylor was himself aware that reciprocal motions would be ineffective at low Reynolds numbers and can be attributed to the first experimental demonstration of the scallop theorem in his classical movie on “Low-Reynolds Number Flows” [3]. In the movie, Taylor uses a mechanical swimmer that consists of an oscillating tail (similar to a boat rudder) and places it first in a basin of water and then in a basin that consists of viscous syrup. While the model swims well in water, in the syrup, it is observed that the swimmer moves backwards and then forwards with no resultant net motion. It was Purcell, who somewhat publicised the theorem in his 1977 paper and rejuvenated this research field. Today, the scallop theorem is relevant in the design of artificial robots [4] for biomedical applications, such as targeted drug delivery and micro-surgery as well as in the development of microfluidic pumps [5] for microelectronics and molecular analysis [6].

For the rest of this section, we start by reviewing some useful properties of the Stokes equations, which we then use in the formal proof of the scallop theorem in the subsequent section. In the following three sections, we discuss three common strategies for circumventing the constraints imposed by the scallop theorem. The first of these strategies is the most intuitive and involves using a non-reciprocal swimming stroke. The latter two involve the addition of new physics, such as inertia and a non-Newtonian rheology to bypass the assumptions under which the scallop theorem holds. For each of these methods, we will present and solve a model to explicitly show that such methods can indeed generate a net flow or force.

1.1 The Stokes equations

In this essay, we will be mostly be concerned with microscopic length scales. This regime can be characterized by a low Reynolds number, where viscous forces are more important than inertia and the Navier Stokes equations reduce to the Stokes equations:

$$\nabla \cdot \boldsymbol{\sigma} = -\mathbf{f}, \quad (1.1)$$

$$\nabla \cdot \mathbf{u} = 0, \quad (1.2)$$

where $\mathbf{f}$ is a body force. In the case of a Newtonian fluid, equations (1.1) and (1.2) simplify to:

$$-\nabla p + \mu \nabla^2 \mathbf{u} = -\mathbf{f}, \quad (1.3)$$

$$\nabla \cdot \mathbf{u} = 0. \quad (1.4)$$

Unlike the complete set of Navier-Stokes equations, these equations are linear and instantaneous due to the lack of time derivatives. An important consequence of the linearity of the Stokes equations is their time-reversibility. If one reverses the forcing for a flow (for example for a flow driven by an external force $\mathbf{F}$, we now consider $\mathbf{F} \rightarrow -\mathbf{F}$), then the direction of the velocity field will reverse ($\mathbf{u} \rightarrow -\mathbf{u}$).

An important solution of the Stokes equations that will be pertinent to this essay is the Stokes flow due to

a point force in three dimensions, which is the solution to:

$$-\nabla p + \mu \nabla^2 \mathbf{u} = -\mathbf{f} \delta(\mathbf{x}), \quad (1.5)$$

$$\nabla \cdot \mathbf{u} = 0. \quad (1.6)$$

This flow field is known as a Stokeslet and can be written in terms of the Oseen tensor $\mathbf{J}$ as $\mathbf{u} = \mathbf{f} \cdot \mathbf{J}$, where $\mathbf{J}$ is defined as:

$$\mathbf{J} = \frac{1}{8\pi\mu} \left(\frac{\mathbf{I}}{r} + \frac{\mathbf{x}\mathbf{x}}{r^3} \right), \quad (1.7)$$

and $r = |\mathbf{x}|$.

1.2 The reciprocal theorem

An important theorem about Stokes flows that will be relevant in this essay is the reciprocal theorem. For Newtonian fluids, the reciprocal theorem can be physically understood as a principle of virtual work (the work done by one flow on another is vice versa), whereas mathematically it can be understood as a form of Green's second identity from vector calculus. In this essay, we will be interested in calculating the translational speed of a swimmer undergoing some time-periodic surface deformation. By using the reciprocal theorem, we can derive an integral identity that will allow us to obtain the average swimming speed without solving for the entire flow field of the swimmer [7]. In order to subsequently generalise the reciprocal theorem, we will now derive it from first principles.

Let $(\mathbf{u}, \boldsymbol{\sigma})$ be the flow field and stress tensor for a force-free and torque-free swimmer with surface $S(t)$ and let $(\hat{\mathbf{u}}, \hat{\boldsymbol{\sigma}})$ be the velocity field and stress tensor for the same instantaneous swimmer shape moving as a rigid body, so we have the following governing equation:

$$\nabla \cdot \mathbf{u} = \nabla \cdot \hat{\mathbf{u}} = 0, \quad (1.8)$$

$$\nabla \cdot \boldsymbol{\sigma} = \nabla \cdot \hat{\boldsymbol{\sigma}} = 0, \quad (1.9)$$

and the following boundary conditions on $S(t)$:

$$\mathbf{u} = \mathbf{U} + \boldsymbol{\Omega} \wedge \mathbf{x} + \mathbf{u}_s, \quad (1.10)$$

$$\hat{\mathbf{u}} = \hat{\mathbf{U}} + \hat{\boldsymbol{\Omega}} \wedge \mathbf{x}, \quad (1.11)$$

where $\mathbf{u}_s$ is the swimming gait and this can be mathematically defined in the Lagrangian sense as $\partial \mathbf{r}_s / \partial t$, where $\mathbf{r}_s$ is a material point on the surface of the swimmer. By using equation (1.9) we have:

$$\int_V \hat{u}_i \partial_j \sigma_{ij} dV = \int_V u_i \partial_j \hat{\sigma}_{ij} dV = 0, \quad (1.12)$$

where we have used suffix notation to avoid ambiguity. We can perform integration by parts to get:

$$\int_S \hat{u}_i \sigma_{ij} n_j dS - \int_V \sigma_{ij} \partial_j \hat{u}_i dV = \int_S u_i \hat{\sigma}_{ij} n_j dS - \int_V \hat{\sigma}_{ij} \partial_j u_i dV = 0, \quad (1.13)$$

where the normal points out of the surface into the fluid and we have neglected the contribution of the surface at infinity to the surface integrals, which is justified provided the velocity and stress fields decay sufficiently fast at infinity. We can use the symmetry of the stress tensor to replace $\nabla \mathbf{u}$ by its symmetric part, which we call $\dot{\boldsymbol{\gamma}}/2$ and then use incompressibility (equation (1.8)) to replace the stress tensor with just its deviatoric component. This gives:

$$\int_S \hat{\mathbf{u}} \cdot \boldsymbol{\sigma} \cdot \mathbf{n} dS - \frac{1}{2} \int_V \boldsymbol{\tau} : \dot{\boldsymbol{\gamma}} dV = \int_S \mathbf{u} \cdot \hat{\boldsymbol{\sigma}} \cdot \mathbf{n} dS - \frac{1}{2} \int_V \hat{\boldsymbol{\tau}} : \dot{\boldsymbol{\gamma}} dV = 0, \quad (1.14)$$

where $\mathbf{A} : \mathbf{B} = A_{ij}B_{ij}$ denotes a double contraction. Equation (1.14) can be viewed as a generalisation of the reciprocal theorem. Using the boundary condition (1.11), we can rewrite the first integral on the left as:

$$\int_S \hat{\mathbf{u}} \cdot \boldsymbol{\sigma} \cdot \mathbf{n} dS = \hat{\mathbf{U}} \cdot \mathbf{F} + \hat{\boldsymbol{\Omega}} \cdot \mathbf{G} = 0, \quad (1.15)$$

where $\mathbf{F}, \mathbf{G}$ are the force and torque on the swimmer from the fluid and the last inequality follows from the fact that the swimmer is assumed to be force-free and torque-free. Then the left hand side of equation (1.14) gives:

$$\int_V \boldsymbol{\tau} : \hat{\boldsymbol{\gamma}} dV = 0. \quad (1.16)$$

Using the boundary condition on the swimmer (equation (1.11)) we can rewrite the surface integral on the right hand side of equation (1.14) as:

$$\int_S \mathbf{u} \cdot \hat{\boldsymbol{\sigma}} \cdot \mathbf{n} dS = \mathbf{U} \cdot \hat{\mathbf{F}} + \boldsymbol{\Omega} \cdot \hat{\mathbf{G}} + \int_S \mathbf{u}_s \cdot \hat{\boldsymbol{\sigma}} \cdot \mathbf{n} dS, \quad (1.17)$$

where $\hat{\mathbf{F}}, \hat{\mathbf{G}}$ are the force and torque on the rigid-body from the fluid. Then using equation (1.14), we obtain the integral identity,

$$\mathbf{U} \cdot \hat{\mathbf{F}} + \boldsymbol{\Omega} \cdot \hat{\mathbf{G}} = - \int_S \mathbf{u}_s \cdot \hat{\boldsymbol{\sigma}} \cdot \mathbf{n} dS + \frac{1}{2} \int_V \hat{\boldsymbol{\tau}} : \hat{\boldsymbol{\gamma}} dV. \quad (1.18)$$

We now consider the case of a Newtonian fluid so we let $\boldsymbol{\tau} = \mu \dot{\boldsymbol{\gamma}}$ and $\hat{\boldsymbol{\tau}} = \mu \hat{\dot{\boldsymbol{\gamma}}}$. Then inserting these expressions for the deviatoric stress and using $\dot{\boldsymbol{\gamma}} : \hat{\boldsymbol{\gamma}} = \hat{\dot{\boldsymbol{\gamma}}} : \hat{\boldsymbol{\gamma}}$ in equation (1.14), we retrieve the classical reciprocal theorem for a Newtonian fluid (with no body force):

$$\int_S \hat{\mathbf{u}} \cdot \boldsymbol{\sigma} \cdot \mathbf{n} dS = \int_S \mathbf{u} \cdot \hat{\boldsymbol{\sigma}} \cdot \mathbf{n} dS. \quad (1.19)$$

We remark that equation (1.19) is independent of equation (1.16) and thus does not rely on the swimmer being force-free and torque-free. However for the case of a force-free and torque-free swimmer, equation (1.16) for a Newtonian fluid can be used to show that:

$$\int_V \hat{\boldsymbol{\tau}} : \hat{\boldsymbol{\gamma}} dV = 0, \quad (1.20)$$

and so in the Newtonian limit, the integral identity (1.18) becomes:

$$\mathbf{U} \cdot \hat{\mathbf{F}} + \boldsymbol{\Omega} \cdot \hat{\mathbf{G}} = - \int_S \mathbf{u}_s \cdot \hat{\boldsymbol{\sigma}} \cdot \mathbf{n} dS. \quad (1.21)$$

What makes this formula useful is that provided we know how to solve the rigid body problem for pure translations or rotations, we can obtain an integral expression for either $\mathbf{U}$ or $\boldsymbol{\Omega}$ without solving for the flow-field of the swimmer by choosing a suitable $(\hat{\mathbf{u}}, \hat{\boldsymbol{\sigma}})$. However, for the formula to be of any practical use, one must be able to solve for $\hat{\boldsymbol{\sigma}}$ in some time-dependent geometry $S(t)$. However, when $S(t)$ deviates only slightly from a reference surface S_0 with enclosed volume V_0 , one can recast the surface and volume integrals onto S_0 and V_0 respectively by using a Taylor expansion in a technique that is known in the literature as ‘‘Domain Perturbation.’’ We now give a brief overview of how this can be done.

Suppose that the material points on the surface can be expressed as:

$$\mathbf{r}_s = \mathbf{r}_0 + \varepsilon \mathbf{r}_1. \quad (1.22)$$

Then the boundary condition on $\mathbf{r} = \mathbf{r}_s$ is given equation (1.10):

$$\mathbf{u}(\mathbf{r}_s) = \varepsilon \frac{\partial \mathbf{r}_1}{\partial t}(\mathbf{r}_0, t) + \mathbf{U} + \boldsymbol{\Omega} \wedge (\mathbf{r}_0 + \varepsilon \mathbf{r}_1). \quad (1.23)$$

By Taylor expanding $\mathbf{u}(\mathbf{r}_s) = \mathbf{u}(\mathbf{r}_0 + \varepsilon \mathbf{r}_1)$ about $\mathbf{r} = \mathbf{r}_0$, we can obtain a boundary condition for $\mathbf{u}(\mathbf{r}_0)$ on the reference surface S_0 of the form:

$$\mathbf{u}(\mathbf{r}_0) = \mathbf{U} + \boldsymbol{\Omega} \wedge \mathbf{r}_0 + \mathbf{u}^{S_0}, \quad (1.24)$$

where $\mathbf{u}^{S_0}$ can be thought of as a swimming gait on S_0 and is expressed as:

$$\mathbf{u}^{S_0} = \varepsilon \frac{\partial \mathbf{r}_1}{\partial t}(\mathbf{r}_0, t) + \boldsymbol{\Omega} \wedge \varepsilon \mathbf{r}_1 - \varepsilon (\mathbf{r}_1 \cdot \nabla) \mathbf{u}(\mathbf{r}_0) + \dots \quad (1.25)$$

Now since we still have $\nabla \cdot \boldsymbol{\sigma} = \mathbf{0}$ in the volume between S_0 and $S(t)$, the swimmer is still force-free and torque free on S_0 , so we can simply move the integrals onto S_0 and V_0 in the integral identity (1.18) to obtain:

$$\mathbf{U} \cdot \hat{\mathbf{F}} + \boldsymbol{\Omega} \cdot \hat{\mathbf{G}} = - \int_{S_0} \mathbf{u}^{S_0} \cdot \hat{\boldsymbol{\sigma}} \cdot \mathbf{n} dS + \frac{1}{2} \int_{V_0} \hat{\boldsymbol{\tau}} : \dot{\boldsymbol{\gamma}} dV. \quad (1.26)$$

We can solve for either $\mathbf{U}$ or $\boldsymbol{\Omega}$ perturbatively if we expand all variables in powers of ε .

1.3 The scallop theorem

Purcell's seminal 1977 paper, titled "Life at low Reynolds number," reviewed the constraints of swimming in an inertia-free world. Here Purcell outlined two crucial and related features of swimming at low Reynolds numbers: rate independence and the infamous scallop theorem. Rate-independence is a consequence of the fact that the Stokes equations are instantaneous. What this means is that swimming at low Reynolds number is a purely geometrical process: the displacement of a swimmer only depends on the sequence of shapes displayed by the swimmer, not the rate at which they occur. Quoting Purcell: "Time doesn't matter. The pattern of motion is the same, whether slow or fast, whether forward or backward in time."

The second is what is now known as the scallop theorem, which depicts the futility of reciprocal deformations. A rather straightforward construction of a reciprocal stroke would consist of a periodic motion, where over one period the motion consists of two different sections. In the first section, the body undergoes some arbitrary deformation, and in the second it undergoes a deformation that is the time-reversal transformation of the motion in the first half. An example of such a motion is depicted in figure 1. Note that due to rate-independence, the time taken for the sequence of shapes to reverse is irrelevant and what only matters is that the same sequence of shapes are the same when viewed forward or backward in time. Purcell used the celebrated example of a scallop that opens and then closes its shell at two different rates to demonstrate an example of a reciprocal surface deformation that would generate no net propulsion. He subsequently suggests that the simplest kind of organism that would be able to swim via a non-reciprocal stroke at low Reynolds number would consist of three links connected by two hinges. This is now known as the Purcell three-link swimmer and examples of reciprocal and non-reciprocal strokes for this swimmer are depicted in figures 1 and 2.

Before proceeding to the next section, we discuss two remarks about the scallop theorem. The first is that non-reciprocal surface deformations require at least two degrees of freedom. This statement can be proved as follows [8]. Let the single degree of freedom be called θ and suppose that the motion is periodic with

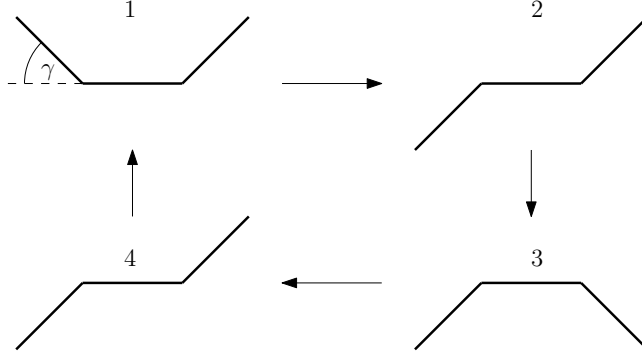


Figure 1: An example of a reciprocal stroke of the Purcell Three-Link Swimmer

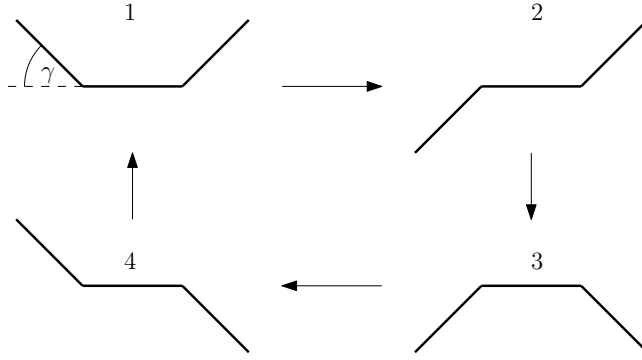


Figure 2: An example of a non-reciprocal stroke of the Purcell Three-Link Swimmer

period T , so that $\theta(0) = \theta(T)$. By the linearity of the Stokes Equations, the swimming speed, $\mathbf{U}$ must be linear in $\dot{\theta}$, so we have:

$$\mathbf{U} = \dot{\theta} \mathbf{F}(\theta). \quad (1.27)$$

If we define $\mathbf{G}(\theta)$ so that $\mathbf{F}(\theta) = \mathbf{G}'(\theta)$, then $\mathbf{G}(\theta)$ is a primitive or indefinite integral of $\mathbf{F}(\theta)$. We can then show that the time average of $\mathbf{U}$ over a period T , denoted $\langle \mathbf{U} \rangle$ is zero, as shown below:

$$\langle \mathbf{U} \rangle = \frac{1}{T} \int_0^T \mathbf{u} dt = \frac{1}{T} \int_0^T \mathbf{G}'(\theta(t)) \dot{\theta}(t) dt = \frac{1}{T} \int_{\theta(0)}^{\theta(T)} \mathbf{G}(\theta) d\theta = 0, \quad (1.28)$$

where in the penultimate inequality, we use a change of variable to convert an integral in t to an integral in θ and the final equality follows from the periodicity in θ .

Secondly, the converse of the scallop theorem does not hold. Namely, there exist non-reciprocal swimming strokes that lead to no net displacement. An elementary example involves a swimmer with a rigid tail and an ellipsoidal head that sweeps out a cone with an axis that passes through the axis of the swimmer's head, as depicted in figure 3. This example has been adapted from an exercise in the textbook by Stephen Childress [9]. Clearly, this motion is not reciprocal, as reversing the direction of time flips the sign of the angular velocity vector of the tail. However, by exploiting the linearity of the Stokes equations and the symmetries of this geometry, it can be shown that this is in fact a non-swimmer.

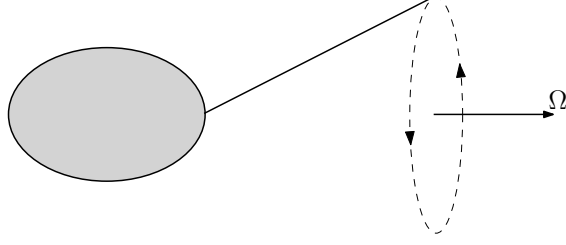


Figure 3: A non-reciprocal non-swimmer

2 A formal proof

As we noted in the previous section, Purcell’s elegant exposition of the Scallop theorem lacks a rigorous formulation as well as a formal proof of the theorem. For example, there is no careful description of what is meant by a “reciprocal motion”, which would require a more precise definition of the surface deformation of the swimmer. Since Purcell, there have been many attempts at a formal proof of the scallop theorem. For example, the much-cited review article by Lauga and Powers [10] as well as the classic textbook by Stephen Childress [9] deal with the specific case where the swimmer only translates but does not rotate. Here we follow a proof by Ishimoto and Yamada [11], which accounts for the rotational motion of the swimmer. One important contribution of their argument is a rigorous mathematical description of the kinematics of a “reciprocal motion.” This requires the introduction of two different frames of reference, which is the starting point of this section. The proof then follows by manipulating an integral identity that is obtained by considering a force balance on the swimmer.

2.1 Kinematics

The crux of their proof lies in the introduction of a virtual swimmer that undergoes the same time-periodic surface deformation as that of the real swimmer except that it is in a vacuum, and thus is tethered—it cannot translate nor rotate in the absence of any fluid that would otherwise exert a force or torque on it. Hence, the virtual swimmer conserves its linear and angular momentum. We can define coordinates with respect to this virtual swimmer, and we can similarly attach a body frame that moves with the real swimmer. Furthermore, we can enforce the condition that the real swimmer and the virtual swimmer coincide at $t = 0$. Then the two frames of reference are related by an affine transformation. However, if we introduce Lagrangian coordinates to label material points on the surface in the two frames, then the Lagrangian coordinates are related by a linear map, namely a rotation matrix. Using the notation of Ishimoto and Yamada, we consider a point $\mathbf{a}$ on the surface of both swimmers at $t = 0$, when both swimmers coincide. Then we can define the Lagrangian coordinate $\mathbf{f}(\mathbf{a}, t)$ to label the location of the material point in virtual coordinates at time t , which was initially at $\mathbf{a}$ at $t = 0$ on the surface of both swimmers (subsequently referred to as a virtual coordinate) and similarly define $\tilde{\mathbf{f}}(\mathbf{a}, t)$ for the real swimmer (body coordinate) (see figure 4). Thus we have for some rotation matrix $\mathbf{R}(t)$:

$$\tilde{\mathbf{f}}(\mathbf{a}, t) = \mathbf{R}(t) \cdot \mathbf{f}(\mathbf{a}, t), \quad (2.1)$$

where $\mathbf{R}(0) = \mathbf{I}$ and $\mathbf{I}$ is the 3×3 identity matrix, as we have enforced the condition that the real and virtual swimmers coincide at $t = 0$.

If the real swimmer rotates with an angular velocity $\boldsymbol{\Omega}(t)$, then the matrix $\mathbf{R}(0) = \mathbf{I}$ satisfies the differential equation:

$$\dot{\mathbf{R}} = \boldsymbol{\Omega} \wedge \mathbf{R}, \quad (2.2)$$

and to avoid any ambiguity, this can also be written in suffix notation as $\dot{R}_{ij} = \varepsilon_{imn} \Omega_m R_{nj}$. This equation follows from considering a basis vector $\mathbf{e}$, that is constant in a rotating frame, that rotates with angular velocity $\boldsymbol{\Omega}$, relative to an inertial frame. Then it is fixed in the rotating frame but has a non-zero time derivative in the inertial frame: $\dot{\mathbf{e}} = \boldsymbol{\Omega} \wedge \mathbf{e}$. It will subsequently be useful to note that the evolution equation for $\mathbf{R}(t)$ can be rewritten as follows:

Lemma 2.1.

$$\dot{R}_{ij} = R_{ik} \Gamma_{kj}, \quad (2.3)$$

where Γ is a second rank tensor which can be defined as:

$$\Gamma_{kj} = -\varepsilon_{kjq} (R^{-1})_{qm} \Omega_m. \quad (2.4)$$

The proof of this lemma is straightforward and follows from manipulating equation (2.2).

Proof. Start from equation (2.2) in suffix notation:

$$\dot{R}_{ij} = \varepsilon_{imn} \Omega_m R_{nj}. \quad (2.5)$$

Then use the definition of the determinant of a 3×3 matrix in terms of the Levi-Civita tensor [12] and the fact that $\mathbf{R}$ is orthogonal, so $\det(\mathbf{R}) = \det(\mathbf{R}^T) = 1$ to obtain the following:

$$\det(R^T) \varepsilon_{ijk} = \varepsilon_{pqr} R_{ip} R_{jq} R_{kr}. \quad (2.6)$$

This gives the following equation for ε_{ijk} :

$$\varepsilon_{ijk} = \varepsilon_{pqr} R_{ip} R_{jq} R_{kr}, \quad (2.7)$$

which we can insert into (2.5) to get:

$$\dot{R}_{ij} = \varepsilon_{pqr} R_{ip} R_{mq} R_{nr} \Omega_m R_{nj} \quad (2.8)$$

$$= \varepsilon_{pqr} R_{ip} R_{mq} \Omega_m \delta_{rj} \quad (2.9)$$

$$= -R_{ip} \varepsilon_{pqj} (R^{-1})_{qm} \Omega_m \quad (2.10)$$

$$= R_{ip} \Gamma_{pj}, \quad (2.11)$$

where we have used the fact that the Levi-Civita tensor is totally antisymmetric and that $\mathbf{R}$ is orthogonal ($\mathbf{R}\mathbf{R}^T = \mathbf{I}$) to obtain equation (2.10). $\square$

It is the simple relationship between the body and the virtual coordinates in equation (2.1) that enables us to derive expressions for the position and orientation of the real swimmer in terms of the virtual coordinates (relative to the virtual swimmer), which will be a crucial ingredient of the proof of the scallop theorem.

The introduction of a virtual swimmer is a classic technique in the fluid mechanics literature that has been applied in numerous previous studies. For example, Yariv [13] used the same technique of introducing two frames of reference, but did not incorporate the rotational motion of the swimmer. We also note that this technique is somewhat reminiscent of a textbook-style proof of the Reynolds transport theorem, where one moves from the Eulerian frame to the Lagrangian frame where the material volume is fixed in time [14].

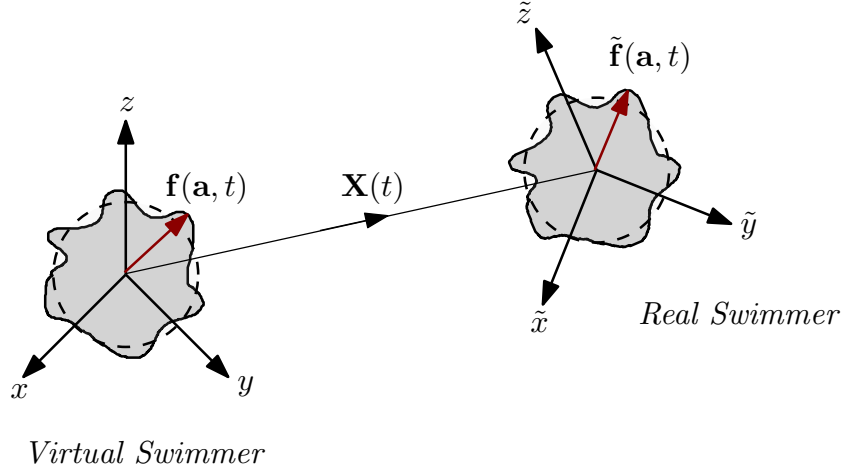


Figure 4: A schematic depicting the body and virtual frame, attached to the real and virtual swimmer, respectively.

The deformation of the swimmer (sometimes called the swimming gait) is prescribed in terms of the body coordinates of the virtual swimmer, denoted $\mathbf{u}'$ and can be defined in a Lagrangian sense as:

$$\mathbf{u}' = \frac{\partial \mathbf{f}}{\partial t}. \quad (2.12)$$

Then the swimming gait for the real swimmer, denoted $\tilde{\mathbf{u}}'$, is linearly related to the swimming gait of the virtual swimmer by construction: $\tilde{\mathbf{u}}' = \mathbf{R}\mathbf{u}'$. The boundary condition for the real swimmer on the surface can be written as:

$$\mathbf{u} = \frac{\partial \tilde{\mathbf{f}}}{\partial t} = \mathbf{U} + \boldsymbol{\Omega} \wedge \tilde{\mathbf{f}} + \tilde{\mathbf{u}}', \quad (2.13)$$

where we have used equations (2.1)-(2.2) and defined $\mathbf{U} = d\mathbf{X}/dt$, where $\mathbf{X}(t)$ is the centre of mass of the real swimmer. Thus, the boundary velocity can be understood as a linear superposition of the flow due to the swimming gait of the real swimmer and the flow resulting from the rigid body motion of the swimmer.

2.2 Forces

In the absence of inertia, a force-free swimmer will experience a thrust and a drag force, and it is the balance of these forces that leads to locomotion. The hydrodynamic drag force is a result of the rigid-body motion of the real swimmer, which is due to the first two terms of its velocity on its surface in equation (2.13). To be consistent with Ishimoto and Yamada, we denote this force as $\mathbf{F}_{\text{TR}}$. There will also be a couple exerted on the real swimmer as a result of its rigid-body motion, and this we call $\mathbf{G}_{\text{TR}}$. By the linearity of the Stokes Equations, we expect $\mathbf{F}_{\text{TR}}$ and $\mathbf{G}_{\text{TR}}$ to be linearly related to $\mathbf{U}$ and $\boldsymbol{\Omega}$ and so we can write:

$$\begin{pmatrix} \mathbf{F}_{\text{TR}} \\ \mathbf{G}_{\text{TR}} \end{pmatrix} = \mathbf{K} \begin{pmatrix} \mathbf{U} \\ \boldsymbol{\Omega} \end{pmatrix}, \quad (2.14)$$

where $\mathbf{K}$ is a symmetric, positive-definite rank-two tensor that only depends on the instantaneous shape of the swimmer $S(t)$ [15].

There will also be a force and torque on the swimmer due to its surface deformation (the last term in equation (2.13)), denoted $\mathbf{F}_{\text{D}}$ and $\mathbf{T}_{\text{D}}$, respectively. Once again, we can exploit the linearity of the Stokes

Equations to deduce that this force and torque must depend linearly on the surface deformation, which may not be uniform at all material points on the surface. This gives the following equation:

$$\begin{pmatrix} \mathbf{F}_D \\ \mathbf{G}_D \end{pmatrix} = \int_{\tilde{S}(t)} d\tilde{S} \begin{pmatrix} \tilde{\mathbf{M}}_u \tilde{\mathbf{u}}' \\ \tilde{\mathbf{M}}_\Omega \tilde{\mathbf{u}}' \end{pmatrix}, \quad (2.15)$$

where $\tilde{\mathbf{M}}_u$ and $\tilde{\mathbf{M}}_\Omega$ are second rank tensors that depend only on $\tilde{S}(t)$, which is the instantaneous shape of the real swimmer at time t .

Then using the fact that the swimmer is force-free and couple-free gives the following equation:

$$\begin{pmatrix} \mathbf{F}_{TR} \\ \mathbf{G}_{TR} \end{pmatrix} + \begin{pmatrix} \mathbf{F}_D \\ \mathbf{G}_D \end{pmatrix} = \begin{pmatrix} \mathbf{0} \\ \mathbf{0} \end{pmatrix}. \quad (2.16)$$

We can insert equations (2.14) and (2.15) into equation (2.16) and invert the matrix $\mathbf{K}$ to get an equation for $\mathbf{U}$ and $\mathbf{\Omega}$ as a function of the surface deformation:

$$\begin{pmatrix} \mathbf{U} \\ \mathbf{\Omega} \end{pmatrix} = -\mathbf{K}^{-1} \int_{\tilde{S}(t)} d\tilde{S} \begin{pmatrix} \tilde{\mathbf{M}}_u \tilde{\mathbf{u}}' \\ \tilde{\mathbf{M}}_\Omega \tilde{\mathbf{u}}' \end{pmatrix} = \int_{\tilde{S}(t)} d\tilde{S} \begin{pmatrix} \tilde{\mathbf{M}}_1 \tilde{\mathbf{u}}' \\ \tilde{\mathbf{M}}_2 \tilde{\mathbf{u}}' \end{pmatrix}, \quad (2.17)$$

where we have defined $\tilde{\mathbf{M}}_1 = -\mathbf{K}^{-1} \tilde{\mathbf{M}}_u$ and likewise for $\tilde{\mathbf{M}}_2$.

We note that for the purposes of proving the scallop theorem, we do not need explicit expressions for $\tilde{\mathbf{M}}_1$ and $\tilde{\mathbf{M}}_2$ in terms of the geometry of $\tilde{S}(t)$. However, such expressions can be obtained by appealing to the integral identity, which was derived in the previous section. Starting from equation (1.21), we can use the resistance matrix for the instantaneous geometrical configuration of the swimmer (denoted $\mathbf{P}$) to get:

$$\begin{pmatrix} \mathbf{U} \\ \mathbf{\Omega} \end{pmatrix} = -\mathbf{P}^{-1} \int_{\tilde{S}(t)} d\tilde{S} \begin{pmatrix} \tilde{\mathbf{u}}' \cdot (\hat{\boldsymbol{\sigma}}_U \cdot \mathbf{n}) \\ \tilde{\mathbf{u}}' \cdot (\hat{\boldsymbol{\sigma}}_\Omega \cdot \mathbf{n}) \end{pmatrix}, \quad (2.18)$$

where we have used the linearity of the Stokes equations to decompose $\hat{\boldsymbol{\sigma}} = \hat{\mathbf{U}} \cdot \hat{\boldsymbol{\sigma}}_U + \hat{\mathbf{\Omega}} \cdot \hat{\boldsymbol{\sigma}}_\Omega$ and used the resistance matrix $\mathbf{P}$ defined as:

$$\begin{pmatrix} \hat{\mathbf{F}} \\ \hat{\mathbf{G}} \end{pmatrix} = -\mathbf{P} \cdot \begin{pmatrix} \hat{\mathbf{U}} \\ \hat{\mathbf{\Omega}} \end{pmatrix}, \quad (2.19)$$

where $\hat{\mathbf{F}}$ and $\hat{\mathbf{G}}$ are the forces and torques exerted by the fluid on the swimmer to be consistent with our choice of the outer unit normal in section 1.

2.3 The proof

We finally have the basic ingredients to rigorously formulate the scallop theorem. We first start by providing a careful definition of a reciprocal deformation. We call a surface deformation reciprocal if there exists some smooth function $g(t)$ such that the material points of the virtual swimmer satisfy the equation:

$$\mathbf{f}(t) = \mathbf{f}(g(t)), \quad (2.20)$$

where we require $g(0) = g(T) = 0$, if the reciprocal motion is to start and end at $t = 0$ and $t = T$. Physically $g(t)$ can be thought of as a rate at which the same sequence of shapes is reversed during a reciprocal deformation. Since $g(t)$ can be an arbitrary smooth function (provided that $g(0) = g(T) = 0$), we have accounted for the case, where the same sequence of shapes may be repeated at a different rate. An immediate consequence of equation (2.20) is that the shape of the virtual swimmer, denoted $S(t)$, must

be the same at times t and $g(t)$, so we have $S(t) = S(g(t))$.

We note that Ishimoto and Yamada consider the more general case, where $\mathbf{f}(t) = \mathbf{Q}(t) \mathbf{f}(g(t))$, where $\mathbf{Q}$ is a three dimensional rotational matrix. This would allow for the possibility that the virtual swimmer takes different directions at time t and $g(t)$. However, this would mean that the virtual swimmer must acquire some angular velocity relative to the virtual frame, call this $\boldsymbol{\omega}$ such that $\dot{\mathbf{Q}} = \boldsymbol{\omega} \wedge \mathbf{Q}$, which follows from equation (2.2). However, since the virtual swimmer is in a vacuum, it has nothing to push on, so it does not translate nor rotate. Thus, we must have $\boldsymbol{\omega} = \mathbf{0}$ and $\mathbf{Q} = \mathbf{I}$, which is what Yamada and Ishimoto explicitly derive.

In terms of the real swimmer, we can deduce from equation (2.20) that the material points on this swimmer must satisfy:

$$\tilde{\mathbf{f}}(t) = \mathbf{R}(t) \mathbf{R}^{-1}(g(t)) \tilde{\mathbf{f}}(g(t)). \quad (2.21)$$

However, since the coordinate system attached to the real swimmer translates and rotates with it, we expect that in terms of body coordinates we must have $\tilde{\mathbf{f}}(t) = \tilde{\mathbf{f}}(g(t))$ and so $\mathbf{R}(t) = \mathbf{R}(g(t))$. This physically corresponds to the fact that the orientation of the real swimmer is independent of the rate at which it deforms its surface. A proof of this observation will serve as the starting point of a proof of the scallop theorem. We commence by recasting the integral identity derived in the previous section onto the virtual swimmer.

For any material point $\mathbf{x}$ that lies on the real swimmer, we have $\mathbf{x} = \mathbf{R}(t) \bar{\mathbf{x}}$, where the material point $\bar{\mathbf{x}}$ lies on the virtual swimmer, which follows from the linear transformation that relates the body frame and the virtual frame from equation (2.1). Thus, equation (2.17) can be recast onto the virtual swimmer as:

$$\begin{pmatrix} \mathbf{U}(t) \\ \boldsymbol{\Omega}(t) \end{pmatrix} = \mathbf{R}(t) \int_{S(t)} dS \begin{pmatrix} \mathbf{M}_u(t) \mathbf{u}'(t) \\ \mathbf{M}_\Omega(t) \mathbf{u}'(t) \end{pmatrix}, \quad (2.22)$$

where we have used $\tilde{\mathbf{u}}' = \mathbf{R} \mathbf{u}'$ and defined the second rank tensors $\mathbf{M}_u = \tilde{\mathbf{M}}_1 \mathbf{R}$ and $\mathbf{M}_\Omega = \tilde{\mathbf{M}}_2 \mathbf{R}$, which depend only on the shape of the virtual swimmer $S(t)$.

Now we perform a change of variable in the above integral with $t' = g(t)$. From equation (2.20) and the chain rule, we have $\mathbf{u}'(t) = g'(t) \mathbf{u}'(g(t))$. Moreover, since $\mathbf{M}_u$ and $\mathbf{M}_\Omega$ are only a function of the shape of the virtual swimmer $S(t)$, we have $\mathbf{M}_u(t) = \mathbf{M}_u(t')$ and $\mathbf{M}_\Omega(t) = \mathbf{M}_\Omega(t')$, which follows from $S(t) = S(g(t))$. This gives:

$$\begin{pmatrix} \mathbf{U}(t) \\ \boldsymbol{\Omega}(t) \end{pmatrix} = g'(t) \mathbf{R}(t) \int_{S(t')} dS \begin{pmatrix} \mathbf{M}_u(t') \mathbf{u}'(t') \\ \mathbf{M}_\Omega(t') \mathbf{u}'(t') \end{pmatrix}. \quad (2.23)$$

We notice that we can rewrite the integral above in terms of $\mathbf{U}(t')$ and $\boldsymbol{\Omega}(t')$ using equation (2.22), which gives us the following set of equations that relate $\mathbf{U}(t)$ and $\mathbf{U}(t')$ as well as $\boldsymbol{\Omega}(t)$ and $\boldsymbol{\Omega}(t')$:

$$\mathbf{U}(t) = g'(t) \mathbf{R}(t) \mathbf{R}^{-1}(t') \mathbf{U}(t'), \quad (2.24)$$

$$\boldsymbol{\Omega}(t) = g'(t) \mathbf{R}(t) \mathbf{R}^{-1}(t') \boldsymbol{\Omega}(t'). \quad (2.25)$$

In order to conclude the proof, we need to show that the matrix $\mathbf{R}(t)$ is rate-independent ($\dot{\mathbf{R}}(t) = \mathbf{0}$). To accomplish this, we use the differential equation satisfied by $\mathbf{R}(t)$ in terms of the tensor $\boldsymbol{\Gamma}(t)$ that was derived in Lemma 2.1. We start from the definition of $\boldsymbol{\Gamma}(t)$ in suffix notation:

$$\Gamma_{ij}(t) = -\varepsilon_{ijk} (R_{km}^{-1} \Omega_m)(t). \quad (2.26)$$

Similarly, at $t' = g(t)$, we have:

$$\Gamma_{ij}(g(t)) = -\varepsilon_{ijk} R_{km}^{-1}(g(t)) \Omega_m(g(t)), \quad (2.27)$$

$$= -\frac{1}{g'(t)} \varepsilon_{ijk} R_{km}^{-1}(t) \Omega_m(t), \quad (2.28)$$

where in the final line, we have used equation (2.25). Combining equations (2.26) and (2.28), we have:

$$\Gamma_{ij}(t) = g'(t) \Gamma_{ij}(g(t)). \quad (2.29)$$

An application of the chain rule and lemma 2.1 gives:

$$\frac{d}{dt} \mathbf{R}(g(t)) = g'(t) \dot{\mathbf{R}}(g(t)) = g'(t) \mathbf{R}(g(t)) \mathbf{\Gamma}(g(t)). \quad (2.30)$$

Using equation (2.29), we notice that $\mathbf{R}(g(t))$ satisfies the same differential equation as $\mathbf{R}(t)$ from lemma 2.1:

$$\frac{d}{dt} \mathbf{R}(g(t)) = \mathbf{R}(g(t)) \mathbf{\Gamma}(t), \quad (2.31)$$

and since they also satisfy the same boundary condition $\mathbf{R}(0) = \mathbf{R}(g(0)) = \mathbf{I}$, we have by the uniqueness of solutions to first order linear ordinary differential equations that $\mathbf{R}(g(t)) = \mathbf{R}(t)$ for all $t \geq 0$. Now, we can go back and use this result to simplify equations (2.24) and (2.25) to get:

$$\mathbf{U}(t) = g'(t) \mathbf{U}(t'), \quad (2.32)$$

$$\mathbf{\Omega}(t) = g'(t) \mathbf{\Omega}(t'). \quad (2.33)$$

If we recall that we defined $\mathbf{U} = d\mathbf{X}/dt$, where $\mathbf{X}$ is the centre of mass of the real swimmer, then we can rewrite equation (2.32) using the chain rule as:

$$\frac{d\mathbf{X}}{dt}(t) = \frac{d}{dt} (\mathbf{X}(g(t))). \quad (2.34)$$

Recall that by the definition of reciprocal we had $g(0) = g(T) = 0$, and so we have an initial condition $\mathbf{X}(0) = \mathbf{X}(g(0))$. We can integrate equation (2.34) and use this initial condition to obtain $\mathbf{X}(t) = \mathbf{X}(g(t))$ for all $t \geq 0$ and in particular $\mathbf{X}(T) = \mathbf{X}(g(T)) = \mathbf{X}(0)$. Thus, there is no net translation of the real swimmer's centre of mass, which concludes the proof of the scallop theorem.

In addition, there are some generalisations of this proof in the literature. Lopez and Lauga [16] extend the proof by Ishimoto and Yamada to fluids with a heterogeneous viscosity (a viscosity that varies with space, $\mu = \mu(\mathbf{x})$). Their proof mirrors that of Ishimoto and Yamada using a generalisation of the reciprocal theorem for heterogeneous viscosities. They then employ a perturbative approach in the limit, where the length scale at which the viscosity varies is much larger than the characteristic length scale of the swimmer and remarkably find that rate independence is preserved at all orders. However, their result must be experimentally verified before it can be practically implemented, for example, in the design of artificial microswimmers in environments with a spatially dependent viscosity.

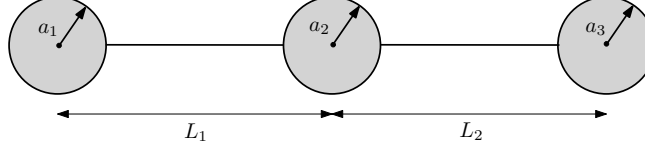


Figure 5: A schematic of the Najafi-Golestanian three sphere swimmer

3 Non-reciprocal kinematics

The scallop theorem is the primary obstacle that must be overcome when attempting to induce some net motion of fluid using a time-periodic deformation. However, the scallop theorem only holds under several constraints: a single swimmer undergoing a reciprocal stroke in a Newtonian, unbounded, inertia-free fluid and each of these constraints provides a mechanism for circumventing the scallop theorem. In this section, we explore one of the most ubiquitous strategies for bypassing the scallop theorem, which is to adopt a non-reciprocal stroke. This can be achieved by using at least two degrees of freedom or by deforming an appendage continuously to generate travelling waves. We begin this section by exploring the former case through a canonical example.

3.1 The three-sphere swimmer

In this section, we consider a specific model for a non-reciprocal swimming stroke and explicitly calculate its swimming speed as a function of a prescribed deformation. The model consists of three spheres that are connected by long and slender rigid rods whose lengths are functions of time. The model is attributed to Najafi and Golestanian [17] and unlike more complicated swimmers, such as the Purcell Three-Link Swimmer, its motion is one-dimensional and thus is more tractable to an analytical approach [18].

A schematic of the geometry can be found in figure 5. The spheres have radii of a_i and the rods have length L_j for $i = 1, 2, 3$, $j = 1, 2$ and the force on sphere i is denoted as f_i . We will be interested in the limit where $a_i/L_j \ll 1$ and we will solve for the swimming speed perturbatively in this limit. Now, if we exert a force of f_i on sphere i , it moves with a velocity of $\tilde{v}_i = f_i/6\pi\mu a_i$ because forces balance in Stokes flows. In the limit $a_i/L_j \ll 1$, each sphere sees another sphere as a Stokeslet with strength equal to the force exerted on the other sphere. Since we are only interested in a leading order calculation, we ignore the hydrodynamic interaction between the spheres (and between the spheres and the rods) and so to leading order the speed of a sphere will be the sum of $\tilde{v}_i$ and velocity field due to the two Stokeslets radiated by the remaining spheres. We can resort to scalars if we assume that $\mathbf{f}_i = f_i \mathbf{e}_x$ and so $\mathbf{v}_i = v_i \mathbf{e}_x$ with the vector $\mathbf{e}_x$ directed along the rod. Then the velocities for each of the spheres can be calculated explicitly as:

$$v_1 = \frac{f_1}{6\pi\mu a_1} + \frac{f_2}{4\pi\mu L_1} + \frac{f_3}{4\pi\mu(L_1 + L_2)}, \quad (3.1)$$

$$v_2 = \frac{f_1}{4\pi\mu L_1} + \frac{f_2}{6\pi\mu a_2} + \frac{f_3}{4\pi\mu L_2}, \quad (3.2)$$

$$v_3 = \frac{f_1}{4\pi\mu(L_1 + L_2)} + \frac{f_2}{4\pi\mu L_2} + \frac{f_3}{6\pi\mu a_3}. \quad (3.3)$$

Since the overall swimmer must be force-free, we have the additional constraint that:

$$f_1 + f_2 + f_3 = 0. \quad (3.4)$$

Hence we can either prescribe the forces on two of the spheres or prescribe the length of the two rods as a function of time ($L_1(t)$ and $L_2(t)$). We will consider the latter case and solve for the forces on the spheres as a function of the length of the rods, which is a prescribed function of time. We further simplify the problem by considering spheres of equal radii $a_1 = a_2 = a_3 = a$, unlike Najafi and Golestanian [18] who solve the general case of unequal radii. The swimming velocity V , is the average translational velocity of the spheres:

$$V = \frac{1}{3}(v_1 + v_2 + v_3) = \frac{1}{12\pi\mu} \left(f_1 \left(\frac{1}{L_1 + L_2} - \frac{1}{L_2} \right) + f_3 \left(\frac{1}{L_1 + L_2} - \frac{1}{L_1} \right) \right), \quad (3.5)$$

where we have used equation (3.4) to eliminate f_2 in favour of f_1 and f_3 .

Since we will be prescribing the deformation of the rigid rods, it will be useful to find V in terms of only L_i and $\dot{L}_i$ for $i = 1, 2$. We can then use equations (3.1)-(3.3) and equation (3.4) to find f_1 and f_3 in terms of the length of speeds of the rods in the limit $a/L \ll 1$ to get:

$$f_1 = -2\pi\mu a(2\dot{L}_1 + \dot{L}_2), \quad (3.6)$$

$$f_3 = 2\pi\mu a(\dot{L}_1 + 2\dot{L}_2), \quad (3.7)$$

which we can insert these expressions into equation (3.5) to get the following expression for V :

$$V = \frac{a}{6} \left(\frac{\dot{L}_2 - \dot{L}_1}{L_1 + L_2} + 2 \left(\frac{\dot{L}_2}{L_1} - \frac{\dot{L}_1}{L_2} \right) + \frac{\dot{L}_1}{L_1} - \frac{\dot{L}_2}{L_2} \right). \quad (3.8)$$

Now we can prescribe some time-periodic deformation of the two rods. Suppose L_1 and L_2 take the form:

$$L_1 = l_1 + d_1(t), \quad (3.9)$$

$$L_2 = l_2 + d_2(t), \quad (3.10)$$

where $d_1(t)$ and $d_2(t)$ are time-periodic. We denote $\langle \rangle$ as an average over this time-period. If the average length of each rod is l_1 and l_2 , then we have $\langle d_1 \rangle = \langle d_2 \rangle = 0$. In order to make analytical progress, we will be interested in the small amplitude, asymptotic limit where $d_i/l_i \ll 1$ for $i = 1, 2$. Inserting equations (3.9) and (3.10) into equation (3.8) and averaging over a period, we find:

$$\langle V \rangle = \frac{a}{3} \langle d_1 \dot{d}_2 \rangle (l_1^{-2} + l_2^{-2} - (l_1 + l_2)^{-2}). \quad (3.11)$$

We can now specify a non-reciprocal harmonic deformation:

$$d_1 = b_1 \cos(\omega t + \phi), \quad (3.12)$$

$$d_2 = b_2 \cos(\omega t), \quad (3.13)$$

where we need $\phi \neq 0, \pi$, so that the oscillations of the rods are not in phase or anti-phase with each other, as this would constitute a reciprocal stroke. Then inserting (3.12) and (3.13) into equation (3.11), we obtain:

$$\langle V \rangle = \frac{ab_1b_2\omega}{6} (l_1^{-2} + l_2^{-2} - (l_1 + l_2)^{-2}) \sin \phi. \quad (3.14)$$

We note here that generalising the above calculation to deal with spheres of different diameters is straightforward, yet tedious [18] and one obtains a swimming speed of:

$$\langle V \rangle = \frac{K}{2} b_1 b_2 \omega \sin \phi, \quad (3.15)$$

where K is a constant defined as:

$$K = \frac{3a_1 a_2 a_3}{(a_1 + a_2 + a_3)^2} (l_1^{-2} + l_2^{-2} - (l_1 + l_2)^{-2}). \quad (3.16)$$

This calculation explicitly demonstrates how a non-reciprocal stroke can be used to circumvent the scallop theorem. Similar analytic results have been obtained for Purcell's three-link swimmer when it undergoes the non-reciprocal motion depicted in figure 2, where one complete cycle consists of the 4 arm strokes in the figure. Purcell himself concluded, based on symmetry arguments, that over one complete cycle, the swimmer will translate in a straight line and without a net rotation. However, Purcell curiously left the direction of translation as "an exercise for the student." This question was investigated almost three decades later by Becker, Coehler, and Stone [19] and their findings indicate that the direction of translation is non-trivial, as it depends on the size of the stroke angle (labelled γ in figure 2) and the lengths of the links. They consider the case where the links (modelled as rigid rods) are slender and use slender body theory to solve for the resistance matrices of the swimmer, which can then be inverted to solve for its swimming speed and angular velocity. This can be numerically integrated forward in time to find the position of the swimmer. For the case where the stroke angle $\gamma = \pi/3$ and the middle link is twice as long as the other links, they find that the swimmer translates to the left in a direction parallel to the middle link in configuration 1 of figure 2 over one period without any net rotation. However, for the same ratio of lengths, large stroke angles (close to π) cause the swimmer to translate in the opposite direction (to the right).

3.2 Comparison with experiments

While the theory presented in the previous section does predict locomotion for the Najafi-Golestanian three-sphere swimmer, it is instructive to ask if this would be practically realised in an experiment and if so, how does the theory compare with experiment. The relative geometric simplicity of the three-sphere swimmer has made it amenable to experimental investigation. We will now discuss an experimental magneto-capillary realisation of this swimmer due to Grosjean et al. [20]. We start with a brief summary of the physics of magneto-capillary swimmers and then subsequently evaluate the calculations in the previous sections with their experimental findings.

A magneto-capillary swimmer consists of ferro-magnetic particles on an air-water interface. The particles are attracted to each other due to capillary effects (in what is known as the Mahadevan-Vella Cheerios effect [21]) as they deform the interface. On the other hand, when an external magnetic field is imposed, the particles experience a repulsive dipole-dipole interaction. The two competing effects cause the particles to "self-assemble" into an equilibrium configuration, which will be a collinear arrangement, provided that the external magnetic field strength exceeds a critical value [22]. For small displacements, this magneto-capillary interaction behaves like a spring-force, and thus when an oscillatory external magnetic field is imposed, the distance between the spheres will also oscillate the same frequency. However, to generate a non-reciprocal motion, one must impose a phase difference, which can be achieved by using beads of different diameters. Grosjean et al. [20] uses a configuration of three spheres of known diameters and then measures the swimming speed (denoted V) for different frequencies of the imposed magnetic field (denoted ω).

In order to verify equation (3.15), they plotted their measured values of the swimming speed V against $W = b_1 b_2 \omega \sin \phi$ and a linear relationship is obtained (which is expected due to the linearity of the stokes equations). This plot can be used to estimate the value of the proportionality constant, which is half the

value of K , as defined in equation (3.16). Subsequently, the authors compute the value of K using the known diameters of the beads and the rest lengths l_1 and l_2 from equation (3.16). This value is found to lie in a 95% confidence interval of the value of K given by the fit on the experimental data, which supports the analysis by Golestainian and Ajdari [18], which was presented in the preceding section.

As with the three-sphere swimmer, there have been macro-scale, experimental realisations of Purcell's three-link swimmer [23]. For example, a study by Or and Gutman [24] used thin aluminium plates as links and joints that were actuated by servo-motors. Low Reynolds number conditions are achieved by using highly viscous silicone oil and sufficiently low angular velocities of the joints. As part of their study, they investigated the dependence of the net translation of the swimmer on its stroke angle for the same gait shown in figure 2. Although for small stroke angles, they also observe only a net translation in the x-direction or parallel to the middle link, they do not recover the reversal of direction for larger stroke angles, which was predicted by slender body-theory. This is partly justified by the mechanical limitation of the servo motors to achieve stroke angles very close to π and so further engineering challenges must be overcome in order to validate the predictions of slender-body theory. Nevertheless, as with the three-sphere swimmer, the crux of such experiments is that they verify the theoretical prediction that there exists a class of non-reciprocal strokes which could lead to net propulsion in the absence of inertia.

3.3 Travelling waves

Unlike the swimmers in the previous sections that have a finite number of degrees of freedom, most biological swimmers use one or more appendages that deform continuously to generate travelling waves. This is not time-reversible, as when viewed in reverse, a travelling wave travels in the opposite direction, thus evading the scallop theorem. There are many different kinds of appendages that are used to generate travelling waves, such as cilia and flagella. Most bacterium, such as *E. coli*, use a passive helical flagellum, which is turned by a molecular motor via a hook on the cell-body. However, some bacteria such as *Cyanobacteria* [25] or spirochetes such as *Borrelia burgdorferi* have no flagella and instead propel themselves via travelling surface waves on their body. On the other hand, eukaryotic flagella, such as those of spermatozoa, are active and consist of molecular motors that are distributed along the length of the filament. These flagella have an internal structure that consists of an arrangement of pairs of microtubules (cylinders made of proteins) that slide past one another. This leads to bending movements that propagate along the flagellum. Moreover, some organisms such as *Paramecium* have their surface covered by many cilia that beat in a coordinated manner to generate what is known as *metachronal waves*. These waves are a result of each cilium undergoing the same deformation, but with a small phase difference, which gives rise to an "envelope" formed by the collective waves.

Two classical models for the generation of travelling waves are the waving sheet [2] and the spherical squirmer [26]. The waving sheet model is attributed to G.I. Taylor and is considered a seminal and classical paper in the field of low Reynolds number locomotion. The model consists of an infinite two-dimensional sheet that undergoes some prescribed waving motion in a viscous fluid. The swimming speed is then solved perturbatively in the long wavelength, small amplitude limit. It is found that for a prescribed transverse wave, the swimming speed is in the opposite direction to the travelling wave, whereas for a purely longitudinal wave, the swimming velocity and travelling wave are in the same direction. The squirmer model can be considered a generalisation of Taylor's waving sheet to a finite spherical swimmer. This can be used as a model for ciliary metachronal waves, which are represented by propagating surface waves. An example of a biological realisation of the squirmer model is the flow-field generated by the green alga *Volvox carteri*. This is a multicellular organism with thousands of flagella on its surface that beat periodically to set up a non-zero time-averaged flow in the fluid. This flow-field can be measured experimentally using particle-image

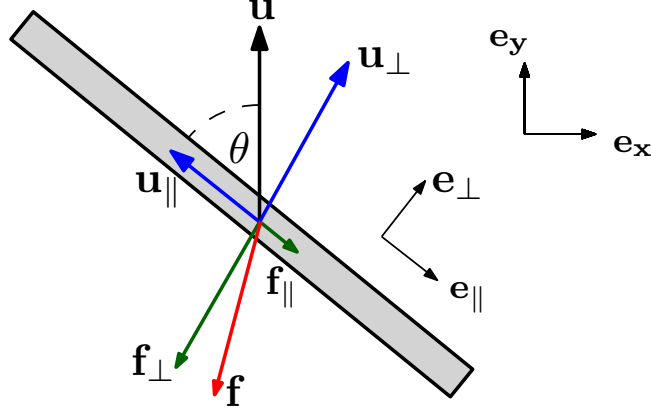


Figure 6: The anisotropic drag on a slender filament

velocimetry and the observed flow is in close agreement [27] with the theoretical prediction of the squirmer model (which must be modified to account for the net force required to hold it in place) .

3.4 Physics of drag based propulsion and optimisation

As we discussed in the previous section, many biological organisms use slender appendages such as cilia or flagella to locomote. Despite the differences between the flagellum of prokaryotes (bacteria) and eukaryotes (such as spermatozoa), the physics behind how these organisms exploit drag to generate propulsion is very similar. This is somewhat remarkable given that these organisms evolved separately many billions of years ago. We now present a physical argument on how propulsion can be achieved for an asymptotically slender filament solely due to viscous drag. In this limit, the drag on the filament can be calculated using resistive-force theory [28], where the force per unit length on the cilium $\mathbf{f}$ is given in terms of the local fluid velocity parallel ($\mathbf{u}_{\parallel}$) and perpendicular to the filament ($\mathbf{u}_{\perp}$) as:

$$\mathbf{f} = -c_{\parallel}\mathbf{u}_{\parallel} - c_{\perp}\mathbf{u}_{\perp}, \quad (3.17)$$

where $c_{\parallel}$ and $c_{\perp}$ are drag coefficients parallel and perpendicular to the filament. For a slender rod, we have $c_{\perp} \approx 2c_{\parallel}$, and so we get a drag anisotropy, where the drag force on the filament is not parallel to the direction of the velocity of the filament. Using the notation and sign conventions in figure 6, we can calculate the force on the filament as:

$$\mathbf{f} = -c_{\perp}u \sin \theta \mathbf{e}_{\perp} + c_{\parallel}u \cos \theta \mathbf{e}_{\parallel}, \quad (3.18)$$

with horizontal component given by:

$$\mathbf{f} \cdot \mathbf{e}_x = (c_{\parallel} - c_{\perp})u \sin \theta \cos \theta. \quad (3.19)$$

It immediately follows from equation (3.19) that no instantaneous horizontal propulsive force would be generated if the drag was isotropic ($c_{\perp} = c_{\parallel}$). However, if we want to generate a net horizontal force with a time periodic deformation, equation (3.19) tells us that we need both $\mathbf{u}$ and θ to change sign over one period. By considering material points on a flagellum over a period, it can be seen that points with locally negative slope ($0 < \theta < \pi/2$) are instantaneously moving upwards ($u > 0$), whereas points with locally positive slope ($-\pi/2 < \theta < 0$) are instantaneously moving downwards ($u < 0$). In both cases, equation

(3.19) gives a horizontal force to the left ($\mathbf{f} \cdot \mathbf{e}_x < 0$), if $c_\perp > c_\parallel$.

Given the ubiquity of travelling waves in biology as a form of propulsion, it is natural to ask if they emerge as the solution to an optimisation problem. While several definitions of optimisation exist in the literature, we shall be concerned with maximising the rate of propulsive or useful work for a fixed amount of energy dissipated in the surrounding fluid (which is equal to the total rate of doing work against the fluid for Stokes flows in the absence of a body force). Using the calculus of variations, it can be shown that optimal planar flagellar kinematics correspond to travelling waves at constant speed [29]. Another relevant optimisation question for slender flagella, first asked by James Lighthill [30], concerns the shape of the most efficient waveform. Assuming that the flagellum is infinite and resistive force theory applies, the optimal waveform has a constant angle between the local tangent and direction of propagation. In three dimensions, this corresponds to a helix, whereas for two dimensions, this is a sawtooth wave, which makes an angle of $\theta \approx \pm 40^\circ$ if $c_\perp/c_\parallel \approx 2$. Some physical intuition for this angle can be gained by appealing to equation (3.19) from the physical argument behind drag-based propulsion. This expression for the total propulsive force is maximised when $\theta = 45^\circ$. However, the total rate at which work is done by the flagellum $\dot{W}$ over one wavelength λ is given by:

$$\dot{W} = - \int_0^\lambda \mathbf{f} \cdot \mathbf{u} ds, \quad (3.20)$$

and using resistive force theory (equation (3.17)), we have:

$$-\mathbf{f} \cdot \mathbf{u} = c_\perp \mathbf{u} \cdot \mathbf{u} + (c_\parallel - c_\perp)(\mathbf{u} \cdot \mathbf{e}_\parallel)^2. \quad (3.21)$$

Since $c_\perp > c_\parallel$, some energy can be saved by increasing $|\mathbf{u} \cdot \mathbf{e}_\parallel|$, which corresponds to decreasing θ , which explains why the optimal angle is less than 45° .

3.5 Artificial microswimmers

The scallop theorem is an important constraint when designing artificial microswimmers, which have several biomedical applications, such as targeted drug delivery and non-invasive surgery [31]. The design of so-called “bio-mimetic swimmers” is often motivated or inspired by the motility strategies employed by biological organisms. A recent development involves artificial swimmers built from soft, organic materials, such as polymers, membranes, or proteins, instead of rigid and hard components. Such soft materials can be highly deformed, which can be exploited to incorporate novel locomotion mechanisms [32].

One example of a bio-mimetic swimmer is a magnetically actuated artificial flagellum. Dreyfus et al. [33] used a linear chain of colloidal magnetic particles linked by DNA in an external oscillating magnetic field. They placed a red-blood cell on one end, which breaks the left-right symmetry, thus giving rise to bending waves that propagate along the filament. A second example involves the fabrication of artificial cilia metachronal waves. In a recent experimental breakthrough, Toonder et al. [34] were able to mimic metachronal waves using $150 \mu m$ long magnetically actuated cilia. They placed an array of cilia made of paramagnetic carbonyl iron in an external magnetic field and were able to distort the field lines, so that at a given time each cilium experienced a different magnetic force, which led to wave-like motion of the cilia.

4 Life at intermediate Reynolds numbers

While reciprocal swimming gaits are futile at low Reynolds numbers because of the scallop theorem, many organisms in the high Reynolds number, Eulerian regime use reciprocal strokes for locomotion. Examples include the flapping of wings by birds and the undulating bodies of fish. Furthermore, many organisms operate at intermediate Reynolds numbers ($Re \sim 0.1 - 1000$), where both viscous and inertial effects must be accounted for. These so-called “Mesoscale” organisms are diverse in both size and swimming mechanisms. Examples include the jet-propulsion of jellyfish [35], the undulatory motions of nematodes [36], and the flapping of pteropods (sea snails) [37]. However, the transition between the different regimes is not well understood. This is important biologically, as many organisms are born into the Stokesian regime due to their relatively small size, but move into the Eulerian regime as they grow. A model organism for this transition is the mollusc *Clione Antartica*, which possesses two modes of locomotion. The first is non-reciprocal and consists of metachronal waves of cilia distributed along the body. The second consists of the reciprocal flapping of its wings. A similar transition can be found in the nymphal mayfly that transitions from rowing to flapping with its gill plates [38]. These organisms raise an interesting fluid-mechanics question on how much inertia is required to evade the constraints of the scallop theorem, so that a reciprocal stroke becomes effective. In this section, we will explore this question and distinguish between the inertia of the fluid and that of the swimmer.

4.1 Reynolds numbers

In the subsequent sections, we will distinguish between the effects of fluid inertia and particle inertia, while further distinguishing the separate contributions to the inertia of the fluid from the unsteady term and advective term in the Navier-Stokes equations. Therefore, we will introduce three different Reynolds numbers to characterise these effects. The oscillatory Reynolds number Re_ω is the ratio of the magnitude of the unsteady term to that of the viscous term. The flapping or frequency Reynolds number Re_f denotes the ratio of the advective term to the viscous term in the Navier-Stokes equations. Finally, the particle Reynolds number Re_p denotes the ratio of the inertia of the particle to the viscous term.

As an example, we can calculate these Reynolds numbers for a spherical particle of density ρ_p , radius a oscillating with frequency ω and amplitude d . The ambient fluid has density ρ , dynamic viscosity μ and kinematic viscosity $\nu = \mu/\rho$. Then $U_0 = d\omega$ is a typical velocity scale and $\rho_p a^3 U_0 \omega$ is a typical magnitude for the particle's inertia (or its rate of change of momentum). Then we obtain the following Reynolds numbers:

$$Re_\omega = \frac{a^2 \omega}{\nu}, \quad (4.1)$$

$$Re_f = \frac{a U_0}{\nu}, \quad (4.2)$$

$$Re_p = \frac{\rho_p a^2 \omega}{\mu}. \quad (4.3)$$

4.2 Fluid inertia

There is experimental evidence that there exists a critical flapping Reynolds number Re_f , above which reciprocal strokes (such as the flapping of a wing) lead to locomotion. A classical experimental study by Childress and Dudley [37] investigated the mollusc *Clione antartica*, which as mentioned previously has two modes of locomotion - one reciprocal and the other non-reciprocal. A more recent experimental study by Webster et al. [39] investigated a different pteropod called *Limacina helicina antarctica* with two similar locomotion modes.

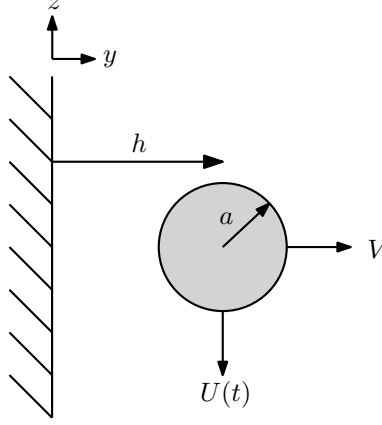


Figure 7: Schematic for an inertial sphere settling near a wall

Both studies found that flapping can only lift the pteropod's weight if the wing motion reaches a critical Reynolds number, which was $Re_f = 35$ in the latter study. Mathematically, this suggests that the transition into the Eulerian regime, where reciprocal strokes can be effective, occurs as a symmetry breaking bifurcation.

In contrast, it has been shown theoretically that the presence of a spatial asymmetry (such as a wall) could lead to a net translational velocity at arbitrarily small values of Re_f for $Re_\omega, Re_p \ll Re_f \ll 1$ and not at a critical value [40]. An elementary example [40] involves a sphere of radius a settling against a wall at a vertical velocity of $U(t)$, as depicted in figure 7. For an inertialess Stokes flow, the sphere does not experience any lateral migration (so its horizontal distance to the wall is a constant). However, sufficiently far from the sphere, inertia will eventually become comparable to viscous forces. A classical calculation [41] using a matched asymptotic expansion shows that for $a \ll h \ll \nu/U$, where U is a typical vertical velocity scale, the leading order lateral migration velocity $V e_y$ is:

$$V = \frac{3}{32} \frac{aU(t)^2}{\nu} > 0. \quad (4.4)$$

Note that since $V > 0$, the formula suggests that the sphere will always migrate away from the wall. We can prescribe a periodic $U(t) = U_0 \cos(\omega t)$ and average V over a period of U to obtain:

$$\frac{\langle V \rangle}{U_0} = \frac{3}{64} Re_f. \quad (4.5)$$

This illustrates how a reciprocal swimming stroke can in fact lead to a net translational velocity for arbitrarily small values of Re_f . The geometrical presence of the wall to break the spatial symmetry is essential to yield net propulsion for this example.

To close this section on fluid inertia, we will discuss the effects of inertia on a dumbbell swimmer, which is composed of two spheres of different radii joined by a light, slender rod. Numerical simulations [42] have demonstrated that when the length of the rod oscillates as some prescribed, time-periodic function, this reciprocal swimmer does in fact locomote. However, what is interesting is that the swimmer undergoes a change in direction as the flapping Reynolds number increases. Here, we refer to the equivalent of the flapping Reynolds number for this system defined as $AR\omega/\nu$, where A and R are the oscillation amplitude and radius of one of the spheres. They found that for $0 < Re_f \leq 20$, the swimmer moves in the direction of the smaller sphere, whereas for $Re_f > 20$, the swimmer reverses direction and now has a translational velocity towards the larger sphere. This can be attributed to a phenomenon that occurs at intermediate

Reynolds numbers, known as “steady streaming.” This is when a time-periodic forcing in the bulk or along an oscillating boundary is enhanced by inertia to produce a steady, non-zero net flow due to the interference of waves

4.3 Particle inertia

In this section, we will consider how the scallop theorem breaks down with a non-zero Re_p . Since we are still enforcing the condition that $Re_f \ll 1$, we are still neglecting the fluid's inertia and thus the Stokes equations and any other results concerning Stokes flows still apply, which is what enables us to make some analytical progress for swimmers in this regime. In this section, we consider a simple model of a heavy inertial dumbbell swimmer, where “heavy,” refers to the fact that we are not ignoring its mass.

The first analytical treatment of this model that we explore is due to Gonzalez-Rodriguez and Lauga [43]. They were interested in the asymptotic regime where $Re_\omega, Re_f \ll Re_p \ll 1$. The crux of their method involves the use of the reciprocal theorem for Stokes flows, which allows them to derive an equation that relates the acceleration and translational velocity of the inertial swimmer to the translational velocity and stress distribution of a massless rigid swimmer of the same geometry that is simply undergoing a rigid-body translation. We will first derive this equation starting from the reciprocal theorem before solving the model. However, we will only consider the case where the inertial swimmer only translates but does not rotate, which means that we do not need to consider a torque balance.

4.3.1 Reciprocal theorem for a heavy inertial swimmer

Since we are still considering the limit, where $Re_f \ll 1$, the reciprocal theorem for Stokes Flows still applies. Let $(\mathbf{u}, \boldsymbol{\sigma})$ be the stress tensor for the flow past a heavy inertial swimmer with boundary condition $\mathbf{u} = \mathbf{U}_p + \mathbf{u}'$ on the boundary of the swimmer, where $\mathbf{u}'$ is the swimming gait or deformational velocity. Let $(\hat{\mathbf{u}}, \hat{\boldsymbol{\sigma}})$ be the velocity and stress tensor of the ambient fluid with $\hat{\mathbf{u}} = \hat{\mathbf{U}}_p$ on the boundary. We can apply the reciprocal theorem to get:

$$\int_{S(t)} \mathbf{u} \cdot \hat{\boldsymbol{\sigma}} \cdot \mathbf{n} dS = \int_{S(t)} \hat{\mathbf{u}} \cdot \boldsymbol{\sigma} \cdot \mathbf{n} dS, \quad (4.6)$$

where $S(t)$ is the instantaneous shape of the swimmer and $\mathbf{n}$ is the unit normal pointing into the fluid. From Newton's second law, we have:

$$m \frac{d\mathbf{u}_G}{dt} = \int_{S(t)} \boldsymbol{\sigma} \cdot \mathbf{n} dS, \quad (4.7)$$

where $d\mathbf{u}_G/dt$ is the acceleration of the centre of mass of the particle and m denotes the mass of the particle. Applying the boundary conditions on the surface give:

$$m \frac{d\mathbf{u}_G}{dt} \cdot \hat{\mathbf{U}}_p - \hat{\mathbf{F}} \cdot \mathbf{U}_p = \int_{S(t)} \mathbf{u}' \cdot \hat{\boldsymbol{\sigma}} \cdot \mathbf{n} dS, \quad (4.8)$$

where we have defined $\hat{\mathbf{F}}$ as the force on the rigid body:

$$\hat{\mathbf{F}} = \int_{S(t)} \hat{\boldsymbol{\sigma}} \cdot \mathbf{n} dS. \quad (4.9)$$

In equation (4.8), we suppose that $\mathbf{u}'$ is prescribed and both $\hat{\mathbf{F}}$ and $\hat{\boldsymbol{\sigma}} \cdot \mathbf{n}$ are known for a particular geometry, then we have an equation for the unknown translational velocity of the swimmer $\mathbf{U}_p$, provided that we can

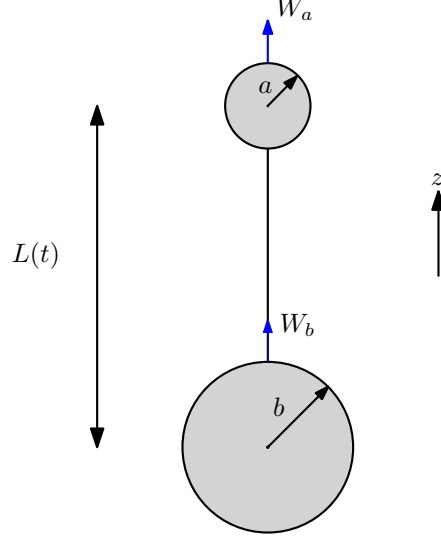


Figure 8: A schematic of the inertial dumbbell swimmer

calculate the acceleration (which, as we will show explicitly, can be worked out from the instantaneous velocities of the swimmer). We also note that by linearity of the Stokes equations for the fluid, $\hat{\mathbf{F}}$ and $\hat{\sigma} \cdot \mathbf{n}$ will be proportional to $\hat{\mathbf{U}}_p$ and so this term cancels out from equation (4.8), as it is arbitrary. This formula can be viewed as an inertial generalisation of the integral identity (1.21), derived in section 1.

4.3.2 An inertial dumbbell swimmer

Our swimmer consists of two heavy spheres of radii a and b (with $a \neq b$ to break the spatial symmetry) with respective velocities W_a and W_b . The spheres have the same density ρ_p and are connected by a light rod of length $L(t)$. We suppose that the length of the rod as a function of time is prescribed and oscillates about its mean value periodically. If the speed of the swimmer (or average translational velocity of the two spheres) is $W(t)$, then we have:

$$W_a(t) = W(t) + w'(t), \quad (4.10)$$

$$W_b(t) = W(t) - w'(t), \quad (4.11)$$

where we have defined the swimming gait or deformational velocity w' as:

$$w' = \frac{1}{2} \frac{dL}{dt}, \quad (4.12)$$

so $W_a - W_b = 2w' = dL/dt$.

We will be interested in averaging the swimming speed W over a period of the deformation, where such an average is denoted $\langle \rangle$. We note that in the absence of particle inertia ($Re_p = 0$), the motion is reciprocal, because we have only one degree of freedom. Hence by the scallop theorem, we should have $\langle W \rangle = 0$. We will now proceed perturbatively in the limit where $\epsilon = a/\langle L \rangle \ll 1$ and a, b are $O(1)$. To apply the reciprocal theorem, we let our dual system be two rigid spheres of the same radii separated by a distance $L(t)$ and each translating at a constant speed $\hat{W}$. We now need to calculate the force $\hat{\mathbf{F}}$. By symmetry $\hat{\mathbf{F}}$ will be in the z -direction and will be equal to $\hat{F}_a + \hat{F}_b$, where $\hat{F}_a$ and $\hat{F}_b$ are the forces on spheres A and B.

In the limit $\epsilon = a/\langle L \rangle \ll 1$, each sphere sees the other as a Stokeslet as with the three-sphere swimmer, which is the next order correction to the velocity of each sphere. Then if the velocity of each sphere is $\hat{W}$ in the z-direction, we have the following expressions for the forces on the spheres by the fluid by using a linear superposition argument, as with the three-sphere swimmer from section 3:

$$\hat{W} = -\frac{\hat{F}_a}{6\pi\mu a} + \frac{3b\hat{W}}{2L}, \quad (4.13)$$

$$\hat{W} = -\frac{\hat{F}_b}{6\pi\mu b} + \frac{3a\hat{W}}{2L}, \quad (4.14)$$

where the minus signs in front of $\hat{F}_a$ and $\hat{F}_b$ appear due to the distinction between the force exerted by the fluid on one of the spheres and the force required to move a sphere at a given velocity. Then we can get the forces exerted on each sphere as:

$$\hat{F}_a = -6\pi\mu a\hat{W} \left(1 - \frac{3b}{2L}\right), \quad (4.15)$$

$$\hat{F}_b = -6\pi\mu b\hat{W} \left(1 - \frac{3a}{2L}\right). \quad (4.16)$$

We note that the above expressions for $\hat{F}_a$ and $\hat{F}_b$ can also be calculated from Faxén's relations for Stokes flows [15]. Since by symmetry, the spheres do not rotate, we can insert our expression for $\hat{F} = \hat{F}_a + \hat{F}_b$ into equation (4.8) (the inertial reciprocal formula) to obtain:

$$m \frac{du_G}{dt} \hat{W} - (\hat{F}_a + \hat{F}_b)W = w'(\hat{F}_a - \hat{F}_b), \quad (4.17)$$

where we have used $\hat{\mathbf{U}}_{\mathbf{p}} = \hat{W}\mathbf{e}_z$, $\mathbf{U}_{\mathbf{p}} = W\mathbf{e}_z$ and $\mathbf{u}' = w'\mathbf{e}_z$ on sphere A and $\mathbf{u}' = -w'\mathbf{e}_z$ on sphere B. We can calculate $m \frac{du_G}{dt}$ as mass times acceleration to get:

$$m \frac{du_G}{dt} = \rho_p \left[\frac{4\pi a^3}{3} \left(\frac{dW}{dt} + \frac{dw'}{dt} \right) + \frac{4\pi b^3}{3} \left(\frac{dW}{dt} - \frac{dw'}{dt} \right) \right]. \quad (4.18)$$

Inserting equation (4.18) into (4.17) yields:

$$\frac{4\pi}{3} \rho_p \left[\frac{dW}{dt}(a^3 + b^3) + \frac{dw'}{dt}(a^3 - b^3) \right] \hat{W} - (\hat{F}_a + \hat{F}_b)W = (\hat{F}_a - \hat{F}_b)w'. \quad (4.19)$$

Consider a sinusoidal displacement $L(t) = \langle L \rangle (1 + A \sin(\omega t))$ and we will solve for W in the limit, where:

$$Re_p = \frac{\rho_p a^2 \omega}{\mu} \ll \epsilon = \frac{a}{\langle L \rangle} \ll 1, \quad (4.20)$$

and we expand W as an asymptotic series in powers of Re_p : $W = W_0 + Re_p W_1 + \dots$. We can then solve for $\langle W \rangle$ order by order. At $O(1)$, we find $\langle \hat{W}_0 \rangle = 0$, which is expected from the scallop theorem as at this order we are not accounting for the inertia of the particles. Thus, the leading order contribution is at $O(Re_p)$, which in terms of $\beta = b/a$ can be expressed as:

$$\langle W \rangle = Re_p a \omega \epsilon^2 \frac{\beta(\beta - 1)(-\beta^2 + 3\beta - 1)}{6(\beta + 1)^2} \left(\frac{A}{a} \right)^2. \quad (4.21)$$

We take note of the following consistency checks. If $\beta = 0$, then $\langle W \rangle = 0$, which is expected, as in this limit, our system reduces to a single sphere oscillating, which is reciprocal. Similarly in the limit $\beta = 1$, we have $\langle W \rangle = 0$, which follows from the up-down symmetry of the system. We also have $\langle W \rangle = 0$ in the limit $\langle L \rangle \rightarrow +\infty$, as there is no hydrodynamic interaction between the spheres in this limit.

One prediction of this model is that the scallop theorem breaks down continuously with particle inertia in the presence of a geometrical asymmetry, as $\langle W \rangle$ depends on a power of Re_p at leading order. This prediction in the absence of fluid inertia agrees with other more recent numerical simulations, such as those of Derr et al. [46]. Another prediction of this model is that $\langle W \rangle > 0$ for $1 < \beta < (3 + \sqrt{5})/2$, so the observed swimming direction is in the direction of the small sphere. However, for $\beta > (3 + \sqrt{5})/2$, the swimming direction reverses. This prediction of the model does not seem to agree with more recent experiments and numerical simulations, which leads us to the following subsection.

4.3.3 Comparison with experimental and numerical simulations

Hubert et al. [44] used a combination of theoretical, numerical, and experimental methods to study the same model of an inertial dumbbell swimmer in an inertia-free fluid ($Re_p = 0$). To be consistent with their notation, we denote the radii of the two spheres as a_1 and a_2 . However, a crucial difference in their study was that they investigated the regime in which the particle Reynolds number Re_p was finite and not asymptotically small. As we shall discuss, it is this region of parameter space that is more amenable to comparisons with experiments. Their theoretical model is obtained by adding an acceleration term to the expression for the velocity of each sphere (such as equations (3.1) to (3.3) for the Najafi-Golestanian swimmer). This gives the following vector equation of motion for the positions of the spheres, which we concatenate into a single vector $\mathbf{x} = (x_1, x_2)$:

$$\frac{\partial \mathbf{x}}{\partial t} = \mathbf{M}(\mathbf{x}) \left[\mathbf{F}(t) - \mathbf{m} \frac{\partial^2 \mathbf{x}}{\partial t^2} \right], \quad (4.22)$$

where $\mathbf{F}(t) = (F_1, F_2)$ is some prescribed force on each sphere, $\mathbf{m}$ is a 2×2 diagonal matrix that consists of m_1 and m_2 , as its diagonal elements. We have also denote the radii of the two spheres as a_1 and a_2 . The matrix $\mathbf{M}$ is a 2×2 matrix that encompasses the hydrodynamic interactions between the spheres (sometimes called a mobility matrix). Analogously to equations (3.1) to (3.3) for the Najafi-Golestanian swimmer, we have:

$$M_{ij} = \begin{cases} (6\pi\mu a_i)^{-1} & i = j \\ (4\pi\mu |x_i - x_j|)^{-1} & i \neq j \end{cases}. \quad (4.23)$$

We note that Hubert et al. consider two cases for how the motion is actuated. The first involves prescribing the forces on the spheres, and in this case Hubert et al. include an additional term in the equation of motion (4.22), which corresponds to a linear Hookean spring force between the spheres. The second case involves prescribing the distance between the two spheres $|\mathbf{x}_1(t) - \mathbf{x}_2(t)| = L + d \sin(\omega t)$. We will only consider the latter case and follow Hubert et al. in solving this equation perturbatively in the limit $a_i/L \ll 1$ (keeping $Re_p = O(1)$) without a spring force to obtain an average swimming speed $\langle U \rangle$ (averaged over a period of the oscillating distance) as:

$$\langle U \rangle = \frac{3d^2}{2} \frac{a_1^2 a_2^2}{(a_1 + a_2)^3 L^2} \frac{\omega(\theta_1 - \theta_2)}{1 + \bar{\theta}^2}, \quad (4.24)$$

where $\theta_i = \tau_i \omega$ is the ratio of a time scale τ_i to the external forcing period for an individual sphere. The sign of $\langle U \rangle$ is chosen so that $\langle U \rangle > 0$ corresponds to motion towards sphere 2 (with position x_2). The time scale τ_i is referred to as a coasting time and is defined as:

$$\tau_i = \frac{m_i}{6\pi\mu a_i}. \quad (4.25)$$

Similarly, we have defined:

$$\bar{\theta} = \frac{(m_1 + m_2)\omega}{6\pi\eta(a_1 + a_2)}, \quad (4.26)$$

which is the same ratio of coasting time to external forcing period for the entire swimmer. The presence of a coasting timescale can be physically understood as a time scale for time taken for a moving spherical particle to come to rest due to the action of a viscous Stokes drag and arises from a scaling relation due to Newton's second law:

$$m \frac{x}{\tau^2} \sim 6\pi\mu a \frac{x}{\tau}. \quad (4.27)$$

The presence of this parameter θ_i in the swimming speed equation $\langle U \rangle$ indicates that the swimming dynamics depend on the ratios of the coasting time and the external forcing time for each sphere. This contrasts with the swimming speed derived by Gonzalez-Rodriguez and Lauga [43], where the swimming speed was predominantly governed by the ratio of the radii of the two spheres.

Hubert et al. verified his theoretical model using numerical simulations and experiments. The numerical simulations used a lattice-Boltzmann method, which in essence relies on simulating the Boltzmann equation from kinetic theory instead of the full Navier-Stokes Equations. This can be less computationally expensive than simulating the full Stokes equations for complicated boundary conditions and can be used to simulate a range of Reynolds numbers [45]. For the spheres, they numerically integrated Newton's equations using a leap-frog method and enforced a no-slip condition on the sphere (using what is known as a "bounce-back" boundary condition for a lattice-Boltzmann simulation). The numerical parameters (the radii and densities of the spheres) are chosen to recover suitable values of the flapping and particle Reynolds numbers, such that the fluid is approximately inertia-free, but the inertia of the particle cannot be neglected.

The experiments were performed using the magneto-capillary swimmers described previously, where the distance between paramagnetic beads on the interface is prescribed by applying an external oscillating magnetic field. Careful measures were taken to ensure that propulsion was only a result of hydrodynamic interaction and not due to external factors, such as capillary forces. Both the stroke amplitude (d) and swimming speed ($\langle U \rangle$) were measured in the experiment, which allowed comparison with the theoretical model using equation (4.24). Furthermore, the assumption that the fluid is approximately inertia-free was verified a posteriori by calculating the fluid Reynolds number based on the measured swimming speed.

We now provide an overview of how their model compares to their experiments and numerical simulations. For small forcing frequencies, they report close agreement between equation (4.24) and the swimming velocity and stroke amplitude measured in experiments. There were discrepancies at higher frequencies, but these could possibly be explained by the importance of fluid inertia or non-linear magneto-capillary interactions that are not captured in the theoretical model. With regard to the numerical simulations, once again there was close agreement between the swimming speed and the forcing frequency, but some deviations were observed for large speeds, which can be attributed to the importance of fluid inertia.

To conclude our discussion of this study, it would be insightful to consider the differences in the predictions of the model proposed by Hubert et al., which accounts for finite particle inertia and that by Gonzalez-Rodriguez and Lauga [43], which considers particle inertia as an infinitesimally small perturbation ($Re_p \ll 1$). One crucial difference is in the swimming direction if both spheres have the same density. The model by Hubert et al. predicts that in this case, the swimmer will always move towards the smaller sphere (in equation (4.24), we have $\langle U \rangle < 0$ for $a_2 > a_1$ using $m_i = 4\pi a_i^3 \rho_p / 3$). However as we have already noted in the previous section, the case $Re_p \ll 1$ predicts a reversal of swimming speed at $\beta = a_2/a_1 = (3 + \sqrt{5})/2$. This does not agree with the experiments performed by Hubert et al., because the spheres have finite inertia that

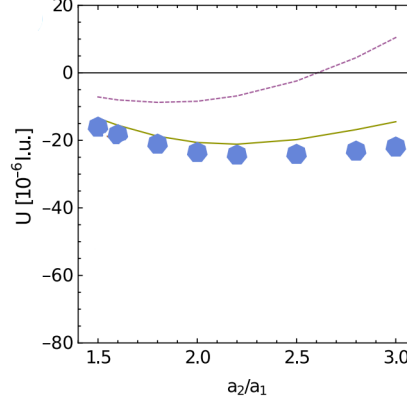


Figure 9: This graph (adapted from [44]) compares the predictions of the average swimming speed (measured in lattice units) as a function of the ratio of the radii of the two spheres from theoretical model (green) and lattice Boltzmann simulations (blue) of Hubert et al. and the perturbative calculation by Gonzalez-Rodriguez and Lauga (purple) [43]

is not asymptotically small and thus the regime considered by Hubert et al. is more relevant for experimental realisations of inertial swimmers. Thus, it is the difference in coasting time scales (that arises due to inertia) for the two spheres that acts as an additional degree of freedom which leads to locomotion for an otherwise reciprocal swimmer.

4.4 Interaction between fluid and particle inertia

Upon comparing the models presented in the preceding sections, it is natural to ask if it would be possible to develop a model that incorporates both particle and fluid inertia. This was attempted by Derr et al. [46], who were motivated by the direction of the swimming speed of dumbbell swimmers. In the previous section, we discussed the study by Hubert et al. [44], which suggested that the swimming direction was always towards the smaller sphere, when accounting only for particle inertia. However, numerical simulations that only account for fluid inertia predict a change in the swimming direction at a critical value of Re_f [42]. Derr et al. aimed to provide a unifying physical explanation in terms of the roles of fluid and particle inertia to explain this switch of direction using a combination of asymptotic analysis and numerical simulations. We first briefly review their setup and then discuss their methods and conclusions.

Their geometrical setup consists of two spheres of radii a_1 and a_2 with the z-axis aligned along the line joining the centres of both spheres. We denote the z-coordinate of each particle as z_1 and z_2 . They consider the case where the distance between the centres of the particles is prescribed as some time-periodic function. We note that from this point onward, we resort to complex notation with the real part being implicitly understood. Thus, we decompose the positions of the particles into an average, steady component and an oscillatory component as $z_j = \bar{z}_j + \hat{z}_j e^{i\omega t}$ ($j = 1, 2$) and denote the inter-particle separation distance as:

$$\zeta = z_2 - z_1 = d + \Delta e^{i\omega t}, \quad (4.28)$$

with d and Δ defined as:

$$d = \bar{z}_2 - \bar{z}_1, \quad (4.29)$$

$$\Delta = \hat{z}_2 - \hat{z}_1. \quad (4.30)$$

We also define the velocity of each sphere as U_k as:

$$U_k = \dot{z}_k = i\omega \hat{z}_k e^{i\omega t}, \quad (4.31)$$

for $k = 1, 2$. They subsequently derive two equations that govern the leading order steady ($\bar{\mathbf{u}}$) and oscillatory ($\hat{\mathbf{u}}$) components of the flow-field in the frame of the swimmer. This is formulated in the perturbative limit $\epsilon = \Delta/a_2 \ll 1$ and $|\Delta| \ll a_1, a_2 \ll d$. Then, they substitute asymptotic expansions for $(\mathbf{u}, p)$ of the form:

$$(\mathbf{u}, p) = \epsilon(\mathbf{u}_1, p_1) + \epsilon^2(\mathbf{u}_2, p_2) + \dots \quad (4.32)$$

into the full Navier-stokes equations. They then pose the following ansatz for $(\mathbf{u}_1, p_1)$ and $(\mathbf{u}_2, p_2)$:

$$(\mathbf{u}_1, p_1) = (\hat{\mathbf{u}}, \hat{p}) e^{i\omega t}, \quad (4.33)$$

$$(\mathbf{u}_2, p_2) = (\bar{\mathbf{u}}, \bar{p}) + (\tilde{\mathbf{u}}, \tilde{p}) e^{i\omega t}. \quad (4.34)$$

The equation for $(\mathbf{u}_1, p_1)$ can be understood as a Fourier series with only the first harmonic to match the $O(\epsilon)$ distance between the particles, which is also oscillating at the same frequency. The ansatz for $(\mathbf{u}_2, p_2)$ in equation (4.34) can be motivated by taking the real part of the advective term $\mathbf{u}_1 \cdot \nabla \mathbf{u}_1$ which has a constant, first harmonic term using the form of $\mathbf{u}_1$ in equation (4.33). However, it can be shown that $(\tilde{\mathbf{u}}, \tilde{p})$ do not contribute to the $O(\epsilon^2)$ swimming speed [46], so we disregard this term. We note that U_k and the swimming speed U are also expanded in this way to get:

$$U_k = \epsilon \hat{U}_k e^{i\omega t}, \quad (4.35)$$

$$U = \epsilon^2 \bar{U}. \quad (4.36)$$

Once we insert the ansatz for $(\mathbf{u}_1, p_1)$ and $(\mathbf{u}_2, p_2)$ into the full Navier-Stokes equations, we obtain the following non-dimensionalised equations for $\hat{\mathbf{u}}$ and $\bar{\mathbf{u}}$ by comparing terms at $O(\epsilon)$ and $O(\epsilon^2)$:

$$(\nabla^2 - iM^2)\hat{\mathbf{u}} = \nabla \hat{p}, \quad (4.37)$$

$$\nabla^2 \bar{\mathbf{u}} = \nabla \bar{p} - \nabla \cdot \bar{\mathbf{R}}, \quad (4.38)$$

where the parameter M^2 is a frequency Reynolds number for this system defined as:

$$M^2 = \frac{\rho \omega a_2^2}{\mu}, \quad (4.39)$$

with ρ denoting the fluid density. The parameter $\bar{\mathbf{R}}$ is defined as:

$$\bar{\mathbf{R}} = -\frac{M^2}{2} \hat{\mathbf{u}} \otimes \hat{\mathbf{u}}^*, \quad (4.40)$$

where $*$ denotes complex conjugate. This parameter emerges from the advective term in the full Navier-Stokes equations and can be interpreted as a ‘‘Reynolds Stress’’ term that is forcing the steady velocity field. We now non-dimensionalise the problem by scaling time with ω^{-1} and lengths with a_2 .

The boundary conditions on $\mathbf{u}$ in the frame moving with the dumbbell are:

$$\mathbf{u} \rightarrow -U \mathbf{e}_z \quad \text{as } |\mathbf{x}| \rightarrow +\infty, \quad (4.41)$$

$$\mathbf{u} = U_k \mathbf{e}_z \quad \text{on } \partial\Omega_k, \quad (4.42)$$

where U is the swimming speed of the dumbbell. We have denoted the surface of sphere k as $\partial\Omega_k$. Since the particles are moving with some prescribed deformation, $\partial\Omega_k$ is not constant in time and so we can move the boundary conditions to $\langle\partial\Omega_k\rangle$ using a Taylor expansion via domain perturbation:

$$\mathbf{u}(z_k) = \bar{\mathbf{u}}(\bar{z}_k) + (z - \bar{z}_k) \frac{\partial \mathbf{u}}{\partial z} \Big|_{z=\bar{z}_k} + \dots = \varepsilon \hat{U}_k e^{it}, \quad (4.43)$$

where the final equality follows from the prescribed swimming gait from equation (4.35). We can then insert our ansatz for $\mathbf{u}$ from equations (4.32)-(4.34) into equation (4.43). Comparing terms at $O(\varepsilon)$ and $O(\varepsilon^2)$, we obtain the following equations for $\hat{\mathbf{u}}$ and $\bar{\mathbf{u}}$ on $\langle\partial\Omega_k\rangle$:

$$\hat{\mathbf{u}} = \hat{U}_k \mathbf{e}_z, \quad (4.44)$$

$$\bar{\mathbf{u}} = \frac{i\hat{U}_k}{2} \frac{\partial \hat{\mathbf{u}}^*}{\partial z}, \quad (4.45)$$

where equation (4.45) follows from the following observation:

$$\text{Re} \left\{ \frac{\partial \mathbf{u}}{\partial z} \Big|_{z=\bar{z}_k} \right\} \text{Re} \{ (z - \bar{z}_k) \} = -\frac{\hat{U}_k \varepsilon^2}{2} \text{Re} \left\{ i \frac{\partial \mathbf{u}^*}{\partial z} \Big|_{z=\bar{z}_k} \right\}, \quad (4.46)$$

and the following non-dimensional result that follows from equations (4.31) and (4.35):

$$z - \bar{z}_k = \hat{z}_k = -i\hat{U}_k \varepsilon. \quad (4.47)$$

We note that when deriving equation (4.43), we have disregarded the $O(\varepsilon^2)$ oscillatory terms, as it can be shown that these terms do not affect the $O(\varepsilon^2)$ swimming speed [46].

Furthermore, we can also further decompose (4.41) and (4.42) into boundary conditions on $\hat{\mathbf{u}}$ and $\bar{\mathbf{u}}$. The boundary conditions at $|\mathbf{x}| \rightarrow +\infty$ are:

$$\hat{\mathbf{u}} \rightarrow 0, \quad (4.48)$$

$$\bar{\mathbf{u}} \rightarrow -\bar{U}. \quad (4.49)$$

Equations (4.37) and (4.38) for $\hat{\mathbf{u}}$ and $\bar{\mathbf{u}}$ can then be solved numerically using a finite element method subject to boundary conditions given by equations (4.44)-(4.45) and (4.48)-(4.49). In order to incorporate the inertia of the particles, they solved Newton's second law for each of the particles, where the inertia of each sphere (mass times acceleration) is equal to the hydrodynamic drag force from the external fluid due to its rigid-body motion and the thrust force associated with its prescribed deformation. Since the governing equations and boundary conditions for the steady component of the solution $(\bar{\mathbf{u}}, \bar{p})$ are linear, we can split the velocity and pressure fields into components driven by the boundary conditions (the swimming gait) and the Reynolds stress: $(\bar{\mathbf{u}}, \bar{p}) = (\bar{\mathbf{u}}_b, \bar{p}_b) + (\bar{\mathbf{u}}_r, \bar{p}_r)$. We can make an analogous decomposition for the swimming speed $\bar{U} = \bar{U}_b + \bar{U}_r$. Then the equation satisfied by $\bar{\mathbf{u}}_b$ is:

$$\nabla^2 \bar{\mathbf{u}}_b = \nabla \bar{p}_b, \quad (4.50)$$

with boundary conditions:

$$\lim_{|\mathbf{x}| \rightarrow +\infty} \bar{\mathbf{u}}_b = -\bar{U}_b \mathbf{e}_z, \quad (4.51)$$

$$\bar{\mathbf{u}}_b = \frac{i\hat{U}_k}{2} \frac{\partial \hat{\mathbf{u}}^*}{\partial z} \text{ on } \langle\partial\Omega_k\rangle. \quad (4.52)$$

On the other hand, the equation satisfied by $(\bar{\mathbf{u}}_r, \bar{p}_r)$ is:

$$\nabla^2 \bar{\mathbf{u}}_r = \nabla \bar{p}_r - \nabla \cdot \mathbf{R}, \quad (4.53)$$

with boundary conditions:

$$\lim_{|\mathbf{x}| \rightarrow +\infty} \bar{\mathbf{u}}_r = -\bar{U}_r \mathbf{e}_z, \quad (4.54)$$

$$\bar{\mathbf{u}}_b = \mathbf{0} \text{ on } \langle \partial \Omega_k \rangle. \quad (4.55)$$

The results of the numerical simulations suggest that if the spheres have the same density as the surrounding fluid and the inertia of the surrounding fluid is accounted for ($M^2 \neq 0$), then the two mechanisms that force the steady velocity field drive motion in opposite directions. In the asymptotic regime, $0 < M^2 \ll 1$, the swimming speed is dominated by the forcing from the boundary conditions ($|\bar{U}| \approx |\bar{U}_b|$), which causes the dumbbell to move towards the smaller sphere. If we increase M^2 so that $M^2 \gg 1$, then the forcing from the Reynolds stress dominates, which causes the sphere to move towards the larger sphere. Thus, their simulations predict a reversal of direction at a critical value of M^2 . In addition, if $M^2 = 0$, so only the particle inertia is accounted for, then it is found that there is no change in direction and the dumbbell moves towards the smaller sphere, thus retrieving the results of Hubert et al. [44].

Despite the progress made by studies such as Derr et al. [46], there are still many experimental challenges associated with verifying the predictions made by theoretical and numerical work on the effects of fluid and particle inertia. For example, at intermediate Reynolds numbers, it can be ambiguous as to which of the three Reynolds numbers (Re_f, Re_ω, Re_p) best characterises a particular system. This is because for experimentally realisable swimmers, the typical length scales, speeds, and frequencies that are observed or used are often of very similar magnitudes, and thus the different Reynolds numbers will also have comparable magnitudes. As a result, it can be difficult to distinguish between the physical effects of the different types of Reynolds numbers in an experiment. This is an obstacle that must be overcome to empirically verify the myriad of results that have emerged purely from theory and numerical simulations.

Moreover, most work on Mesoscale swimming has focused on single, individual swimmers in relatively simple geometries such as a spherical dumbbell or a spherical squirmer [47]. There is limited work on collective locomotion at intermediate Reynolds numbers. This is pertinent to several biological processes, such as “biogenic mixing”, where the flows around Mesoscale swimmers, such as zoo-plankton and brine shrimp, contribute to the large-scale mixing of nutrients in the ocean [48]. However, the study of collective Mesoscale swimming brings its own set of mathematical challenges. Since the Stokes equations no longer apply at intermediate Reynolds numbers, the full Navier-Stokes equations must be solved directly. Consequently, it can be computationally intensive to numerically simulate the non-linear hydrodynamic interactions between a large number of Mesoscale swimmers.

In addition, there are similar questions about how the scallop theorem breaks down in the presence of advection and viscosity gradients. As we have already mentioned in a previous section, a version of the scallop theorem exists for fluids that have a heterogeneous viscosity [16], which may depend on the temperature or concentration of a solute ($\mu = \mu(C, T)$). This generalisation holds, provided that there is no advection of the temperature and concentration fields, so that the steady temperature and concentration fields satisfy Laplace’s equation. An unexplored, yet natural question is how much advection is required to circumvent the scallop theorem and whether it breaks down at a finite amount of advection (as a mathematical bifurcation) or whether the breakdown is continuous. This is particularly relevant in biology as many organisms, such as the green algae *Volvox* live in high Péclet number environments and advection by fluid flow plays a key role in nutrient uptake and waste dispersal [49].

5 Complex fluids

One crucial assumption for the scallop theorem is that it is only valid for a Newtonian fluid, and so one could hope that the theorem could be evaded by the presence of a non-Newtonian or complex fluid. Such fluids are ubiquitous in biology and a notable example is the cervical mucus in the female reproductive tract [50]. Non-Newtonian fluids are often mathematically described by non-linear constitutive relations, and it is these inherent non-linearities that could potentially cause the scallop theorem to break down. These non-linearities can take different physical forms and here we discuss two such cases: normal stress differences due to non-linear elastic effects and the rate-dependence of deformations, which arises due to the shear viscosity depending non-linearly on the shear rate. We first discuss some relevant rheological background.

5.1 Rheological models

Viscoelastic fluids lie somewhere between an elastic solid and a Newtonian Viscous fluid. An important time scale that appears due to the emergence of its elastic solid-like behaviour is that of a relaxation time, denoted λ . An elastic Hookean spring returns to its original length, when it is stretched (within its elastic limit), and so has some memory of its initial configuration, and the timescale λ is some measure of how long a fluid remembers its flow history. For a given flow, the importance of the relaxation time λ depends on how long we are observing the fluid and can be characterised by a non-dimensional number called the Deborah number:

$$De = \frac{\lambda}{T}, \quad (5.1)$$

where T is the time of observation. Thus elastic effects are important for $De \gg 1$.

In order to solve the Stokes equations for a non-Newtonian fluid, one must specify a constitutive equation for the deviatoric stress $\boldsymbol{\tau}$. While there is a diverse variety of complex fluids, we shall focus on two different rheological models, namely the Oldroyd-B and second-order fluid models. The Oldroyd-B model is a standard model for dilute polymer solutions. The stress tensor is divided into a polymeric and Newtonian solvent contribution: $\boldsymbol{\tau} = \boldsymbol{\tau}_p + \boldsymbol{\tau}_s$ with a solvent contribution $\boldsymbol{\tau}_s = \eta_s \dot{\boldsymbol{\gamma}}$ and $\boldsymbol{\tau}_p$ satisfying the constitutive equation:

$$\boldsymbol{\tau}_p + \lambda \overset{\nabla}{\boldsymbol{\tau}}_p = \eta \dot{\boldsymbol{\gamma}}, \quad (5.2)$$

where $\lambda = \eta/G$ is the relaxation time and $\overset{\nabla}{\mathbf{A}}$ denotes the upper convective derivative, defined as:

$$\overset{\nabla}{\mathbf{A}} = \frac{D\mathbf{A}}{Dt} - (\nabla \mathbf{u})^T \cdot \mathbf{A} - \mathbf{A} \cdot (\nabla \mathbf{u}). \quad (5.3)$$

The Oldroyd-B constitutive equation can also be derived using ideas from Kinetic Theory, where a dilute solution of polymer molecules is modelled as a suspension of non-interacting Hookean dumbbells in a Newtonian solvent. Each polymer molecule is modelled as a dumbbell consisting of two beads connected by an infinitely extensible spring, which accounts for the drag experienced by the polymer molecule.

Another relevant constitutive equation is that of a second-order fluid, which can be thought of as a perturbation of a Newtonian fluid for small amounts of elasticity ($De \ll 1$). It is described by:

$$\boldsymbol{\tau} = \eta \dot{\boldsymbol{\gamma}} + \beta_1 \overset{\nabla}{\dot{\boldsymbol{\gamma}}} + \beta_2 \dot{\boldsymbol{\gamma}} \cdot \dot{\boldsymbol{\gamma}}, \quad (5.4)$$

where β_1 and β_2 are material parameters.

5.2 Elastic effects

An additional piece of physics that arises as a result of the inherent microstructure present in viscoelastic fluids is that of normal stress differences. Physically, this can be understood as a consequence of entropic forces due to polymers being stretched in an ambient shear flow, which leads to an apparent tension in the streamlines. We now present a model for a reciprocal two-sphere swimmer in a viscoelastic fluid. Due to the inherent non-linearities present in many non-Newtonian rheologies, it is often necessary to resort to numerical simulations. However, some analytical progress can be made by once again appealing to the reciprocal theorem in the small-amplitude limit to avoid having to calculate the entire flow-field for the non-Newtonian swimmer. We first outline the mathematics behind how this is done in practice and then demonstrate how the reciprocal theorem can be applied to a viscoelastic dumbbell swimmer in an Oldroyd-B fluid. Finally, we discuss a physical mechanism that enables reciprocal swimmers to evade the scallop theorem via the emergence of secondary flows.

5.2.1 A generalisation of the reciprocal theorem

Suppose that the deviatoric stress tensor $\boldsymbol{\tau}$ in the generalised reciprocal theorem (1.14) can be written in the form:

$$\boldsymbol{\tau} = \eta \dot{\boldsymbol{\gamma}} + \mathbf{N}(\mathbf{u}, \boldsymbol{\tau}), \quad (5.5)$$

where $\mathbf{N}$ is a symmetric tensor that depends non-linearly on the velocity and stress fields and can be thought of as a small perturbation from a Newtonian fluid at low Deborah number, such as for a second-order fluid. However, other more general viscoelastic rheologies, such as Oldroyd-B, can also be put into this form if we move into Fourier space and allow η to be complex [51]. We suppose that our auxiliary problem satisfies:

$$\hat{\boldsymbol{\tau}} = \hat{\eta} \hat{\boldsymbol{\gamma}}. \quad (5.6)$$

Then we can calculate the non-Newtonian contribution in the generalised reciprocal theorem (equation (1.14)) from equations (5.5) and (5.6) as:

$$\int_V \hat{\boldsymbol{\tau}} : \hat{\boldsymbol{\gamma}} dV = \hat{\eta} \int_V \hat{\boldsymbol{\gamma}} : \hat{\boldsymbol{\gamma}} dV, \quad (5.7)$$

$$= \frac{\hat{\eta}}{\eta} \int_V \hat{\boldsymbol{\gamma}} : \boldsymbol{\tau} - \mathbf{N} : \hat{\boldsymbol{\gamma}} dV, \quad (5.8)$$

$$= -\frac{\hat{\eta}}{\eta} \int_V \mathbf{N} : \hat{\boldsymbol{\gamma}} dV, \quad (5.9)$$

where the first term in (5.8) is zero by equation (1.16). Then upon inserting equations (5.5) and (5.6) into the generalised integral identity (equation (1.26)) from section 1, we have in the limit of small amplitude deformations:

$$\mathbf{U} \cdot \hat{\mathbf{F}} + \boldsymbol{\Omega} \cdot \hat{\mathbf{G}} = - \int_{S_0} \mathbf{u}^{S_0} \cdot \hat{\boldsymbol{\sigma}} \cdot \mathbf{n} - \frac{\hat{\eta}}{2\eta} \int_{V_0} \mathbf{N} : \hat{\boldsymbol{\gamma}} dV. \quad (5.10)$$

We now consider a more general rheology of the form:

$$\mathcal{A}[\boldsymbol{\tau}] = \eta \mathcal{B}[\dot{\boldsymbol{\gamma}}] + \mathbf{N}[\mathbf{u}, \boldsymbol{\tau}], \quad (5.11)$$

where $\mathcal{A}$ and $\mathcal{B}$ are linear operators in time. We note that a constitutive equation, such as equation (5.11) can be put into the form of equation (5.5) if all functions (and the unknowns $\mathbf{U}, \boldsymbol{\Omega}$) are periodic and can be expanded as a Fourier series [51]. For example:

$$\mathbf{u} = \sum_n \mathbf{u}^{(n)} e^{in\omega t}. \quad (5.12)$$

At each Fourier mode, we obtain the equation of the form:

$$\boldsymbol{\tau}^{(n)} = \eta^*(n) \dot{\boldsymbol{\gamma}}^{(n)} + \mathbf{N}^{(n)}, \quad (5.13)$$

for some complex viscosity $\eta^*(n)$, which depends on the characteristic polynomials of the differential operators $\mathcal{A}$ and $\mathcal{B}$. We can subsequently expand $\mathbf{U}$, $\boldsymbol{\Omega}$, $\mathbf{u}^{S_0}$, $\mathbf{N}[\mathbf{u}, \boldsymbol{\tau}]$ in powers of ε to solve for the swimming speed and angular velocity order by order. For example:

$$\mathbf{U} = \varepsilon \mathbf{U}_1 + \varepsilon^2 \mathbf{U}_2 + \dots \quad (5.14)$$

However, we will often be interested in the mean swimming velocity and angular velocity, which corresponds to the zeroth mode, $\langle \mathbf{U} \rangle = \mathbf{U}^{(0)}$, $\langle \boldsymbol{\Omega} \rangle = \boldsymbol{\Omega}^{(0)}$. It can be shown [52] that at $O(\varepsilon^m)$, the mean swimming speed and angular velocity satisfy the following equation:

$$\langle \mathbf{U}_m \rangle \cdot \hat{\mathbf{F}} + \langle \boldsymbol{\Omega}_m \rangle \cdot \hat{\mathbf{G}} = - \int_{S_0} \langle \mathbf{u}_m^{S_0} \rangle \cdot \hat{\boldsymbol{\sigma}} \cdot \mathbf{n} - \frac{\hat{\eta}}{2\eta^*(0)} \int_{V_0} \langle \mathbf{N}_m \rangle : \hat{\boldsymbol{\gamma}} dV. \quad (5.15)$$

For a time periodic swimming gait, we have $\langle \mathbf{u}^{S_0} \rangle = \mathbf{0}$, and so by equation (5.15), $\mathbf{U}$ and $\boldsymbol{\Omega}$ only depend on the non-linear, non-Newtonian contribution to the stress tensor. This means that for a linearly viscoelastic fluid, such as a linear Maxwell fluid, where we would have $\mathbf{N} = \mathbf{0}$, equation (5.15) implies that $\mathbf{U} = \boldsymbol{\Omega} = \mathbf{0}$ as $\hat{\mathbf{F}}$ and $\hat{\mathbf{G}}$ are arbitrary. Hence, the scallop theorem holds for a linear viscoelastic fluid. However, this is not practically useful because linear viscoelastic models such as the Maxwell model are not physically valid because they are not objective.

Since $\mathbf{N}$ represents the non-linear terms in a specified rheology, we expect it to have no term linear in ε and at leading order we expect $\mathbf{N} = \varepsilon^2 \mathbf{N}_2(\mathbf{u}_1, \boldsymbol{\tau}_1)$. Hence we have $\mathbf{U}_1 = \boldsymbol{\Omega}_1 = \mathbf{0}$ and at leading order for a periodic swimming gait we have:

$$\langle \mathbf{U}_2 \rangle \cdot \hat{\mathbf{F}} + \langle \boldsymbol{\Omega}_2 \rangle \cdot \hat{\mathbf{G}} = - \frac{\hat{\eta}}{2\eta^*(0)} \int_{V_0} \langle \mathbf{N}_2 \rangle : \hat{\boldsymbol{\gamma}} dV. \quad (5.16)$$

Since $\mathbf{N}_2 = \mathbf{N}_2[\mathbf{u}_1, \boldsymbol{\tau}_1]$, if we are to use equation (5.16) we need to solve for $\mathbf{u}_1$. However, from equation (5.13) at $O(\varepsilon)$ we have for each Fourier mode:

$$\boldsymbol{\tau}_1^{(n)} = \eta^*(n) \dot{\boldsymbol{\gamma}}_1^{(n)}, \quad (5.17)$$

and thus we have a Newtonian problem with viscosity $\eta^*(n)$ for the n^{th} Fourier mode, and so the velocity field will be identical to the Newtonian solution for the same geometry [52].

5.2.2 A viscoelastic two-sphere swimmer

We now give an explicit example of how the scallop theorem breaks down in a non-Newtonian fluid. We will consider a two-sphere dumbbell swimmer in an Oldroyd-B fluid, where the distance between the spheres is prescribed to vary sinusoidally. We can then use equation (5.16) to calculate its net translational velocity in the small amplitude limit. Remarkably, an analytical solution exists for the rigid body motion of two spherical particles moving along the line joining their centres at different velocities [54]. The solution is in terms of an infinite series involving Legendre polynomials and is obtained by solving the Stokes equations using a Stokes streamfunction in bipolar spherical coordinates. This can be taken as our auxiliary problem and gives us $\hat{\mathbf{F}}$, $\hat{\boldsymbol{\gamma}}$ and $\mathbf{u}_1$ with $\hat{\mathbf{G}} = \mathbf{0}$. We can obtain an expression for $\mathbf{N}_2$ by expanding the Oldroyd-B constitutive equation in Fourier space. Therefore, the solution to $\mathbf{U}_2$ reduces to numerically evaluating an integral on the right hand side of equation (5.16). An analogous approach can be used to calculate the

translational velocity of a reciprocal swimmer that consists of two rigid spheres of different radii that can rotate as a rigid body in a second order fluid [55].

The physical mechanism that enables reciprocal motion in viscoelastic fluids to escape the constraints of the scallop theorem is due to the presence of normal stress differences. For example, a single sphere oscillating in a viscoelastic fluid generates a secondary flow that draws fluids towards the centre of the sphere along the axis of oscillation [56], which is a result of tension in the streamlines creating a hoop stress. This flow has a non-zero time average and is thus often referred to as a steady streaming flow. The strength of this secondary flow depends on the radii of the sphere, and so when one has a configuration that consists of two spheres of different sizes, the interacting secondary flows could have a non-zero time average and thus lead to a net translational velocity. We also note that this mode of propulsion has been realised experimentally, using magnetically actuated artificial microswimmers that undergo reciprocal swimming gaits in elastic fluids [57].

5.3 Rate dependence

Another strategy for circumventing the scallop theorem is by exploiting the so-called rate-dependence of a generated flow on the swimming gait. For a Newtonian fluid undergoing a Stokes flow, time disappears from the governing equations and boundary conditions. However, in a non-Newtonian fluid, the flow or forces induced by a surface deformation do not just depend on the instantaneous geometry of the surface, but could also depend on the time taken to deform a surface from one configuration to another. This is particularly useful for fluids that are not elastic, such as shear-thinning and shear-thickening fluids, which do not exhibit normal stress differences. In an experimental breakthrough, this method of propulsion has been experimentally realised using a magnetically actuated, single-hinge, micro-scallop swimmer in a shear-thickening and shear-thinning fluid [58]. By implementing a time asymmetric stroke, where the scallop opens and closes at different rates, the two sections of the motion induce different shear rates in the ambient fluid, which leads to different viscosities. Hence, the thrust force on the swimmer, which is due solely to the swimming gait, is different when opening and closing the scallop. As a control, it was verified that a symmetric stroke with equal opening and closing rates leads to no net propulsion, and by direct measurement, they experimentally confirmed that normal-stress differences were negligible.

The variety of non-Newtonian constitutive relations makes it difficult to propose general statements about the impact of a particular rheological characteristic. For example, for an undulatory swimmer, theoretical predictions suggest that the swimming speed is impeded by elasticity [59], but has no effect in a shear-thinning Carreau fluid [60], which does not exhibit elastic effects, such as normal stress differences. These findings have been verified by experiments with the nematode *C. elegans* [61][62]. Further theoretical and experimental work is required to make universal statements on the impact of a particular non-Newtonian characteristic, such as elasticity or a shear-dependent viscosity that do not rely on a particular geometry and thus provide a unifying perspective to the myriad of conclusions that have been drawn by other studies. Moreover, there are experimental challenges when experimenting with biological organisms in non-Newtonian fluids. This is because organisms often change their swimming gait in response to fluid rheology. For example, the green alga *C. reinhardtii* exhibits different beating patterns of its two flagella in a viscoelastic fluid as opposed to a Newtonian fluid [63]. Often, numerical simulations must be performed in conjunction with experiments to decouple the effects of elasticity and the change in swimming kinematics. Whether this change in swimming gait is an active or passive response to the change in rheology is an unresolved problem and an ongoing effort in the research community.

6 Conclusion and discussion

In this essay, we have reviewed the scallop theorem and its profound consequences for generating net flows and forces from periodic surface deformations. We presented a rigorous proof of the theorem and then discussed three different methods to circumvent the constraints of the theorem and their relevance to biology and in the design of artificial swimmers. Dumbbell swimmers have been our mathematical model of choice, due to their amenability to analytical calculations in the small amplitude limit. However, there are other mechanisms for bypassing the scallop theorem that we have not discussed, such as hydrodynamic interactions and thermal noise. In terms of hydrodynamic interactions, two swimmers undergoing the same reciprocal swimming gait with a different phase can locomote [65], since the phase difference can be understood as an additional degree of freedom. With regard to noise, it has been theoretically demonstrated that a two-sphere dumbbell swimmer subject to Brownian noise in a geometrically confined environment can generate a net displacement in a statistical sense (when taking an ensemble average over many realisations of the noise) [66].

While we have mainly focused on how the scallop theorem breaks down, a mathematically interesting question is if it would be possible to incorporate additional physics so that the scallop theorem does in fact hold and so periodic, reciprocal strokes are still futile for exerting net forces on the fluid. In this essay, we have briefly mentioned two such generalisations, namely that for heterogenous viscosity gradients [16] and for linear viscoelastic fluids. However, the latter is not a particularly useful result in practice as linear viscoelastic fluid models are not objective and thus can be rendered as unphysical. There is limited research in this area and understanding environments and conditions where the scallop theorem can be generalised could prove to be useful for engineering artificial microswimmers, if any such environments or conditions are of biological or industrial relevance.

There are several experimental and computational challenges that must be addressed to verify the theoretical work on circumventing the scallop theorem. In terms of experiments, it can possibly be difficult to isolate a phenomenon of interest. We have already alluded to this issue in our discussion of inertia when distinguishing between the effects of particle and fluid inertia in experiments. There are also analogous issues when distinguishing between viscous and elastic effects in complex fluids. Similarly, there are computational challenges when performing numerical simulations with a large number of swimmers. One reason for this is that it becomes computationally demanding to resolve the many-body hydrodynamic interactions, such as the lubrication flow between microswimmers. With current computing capabilities, it is possible to numerically resolve this lubrication flow for about a 1000 swimmers [67], but this figure will have to increase by at least 3 orders of magnitudes to be pertinent to the large number of organisms that are observed in nature. While a significant amount of progress has been made since the conception of the scallop theorem, we have now reached the stage where further developments in experimental and numerical techniques are required to keep up with the theory to make further progress.

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